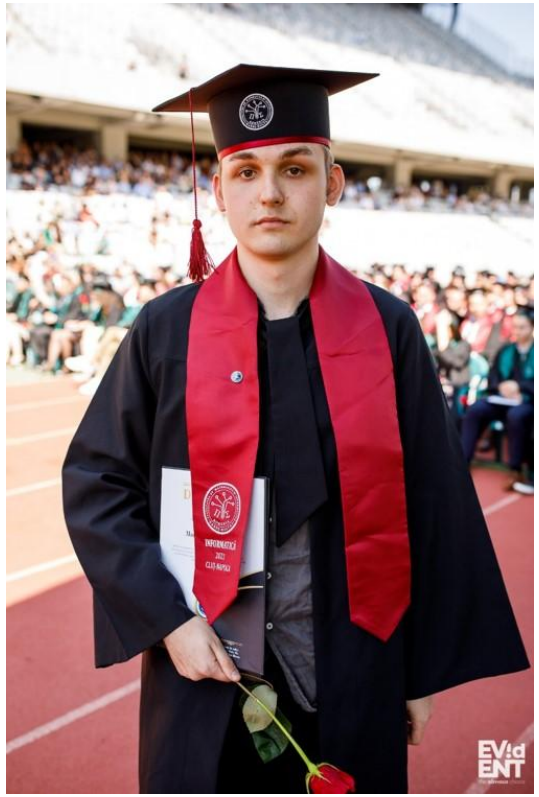
Inteligență Artificială Generativă - LLMs

AI Curs 6 – 31.03.2026

Conținut Curs

1. Neural Networks, CNN Recap
2. Attention, Transformers
3. Generative AI (LLMs)
4. Embeddings

Despre mine



- Student la doctorat în anul III la UBB FMI
- Domenii de interes: Computer Vision, Multitask Models, Image Generation
- alexandru.manole@ubbcluj.ro

Despre voi

- https://www.menti.com/
- cod: **1837 9742**

Menti I

1837 9742

Menti I

Îmi place să...

rezolv probleme de AI

fac sport

mănânc ficăței

rezolv probleme folosind AI

ies cu colegii

mă uit la filme \ seriale

Strongly disagree

Strongly agree

Menti II

Mi s-a părut dificil laboratorul:

Lab 1 - Algoritmi

Lab 2 - Preprocesari

Lab 3 - OCR

Lab 4 - Clasificare + Detecție

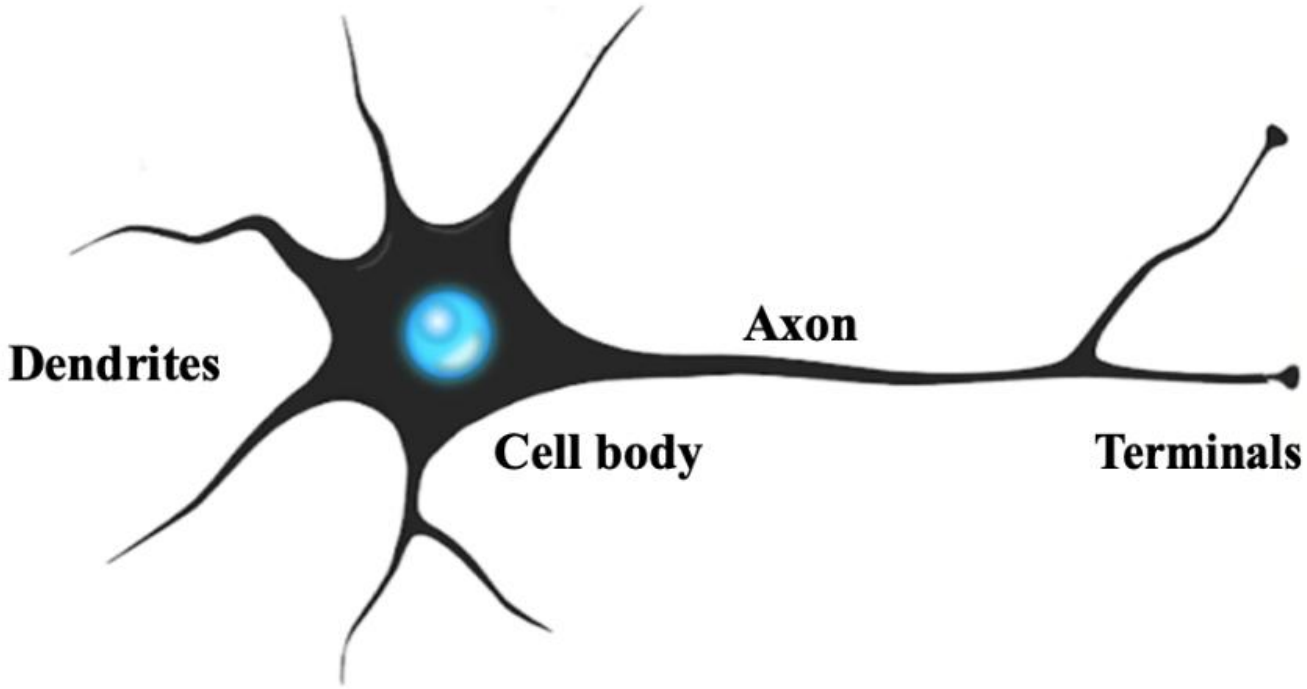
Lab 5 - Regresie

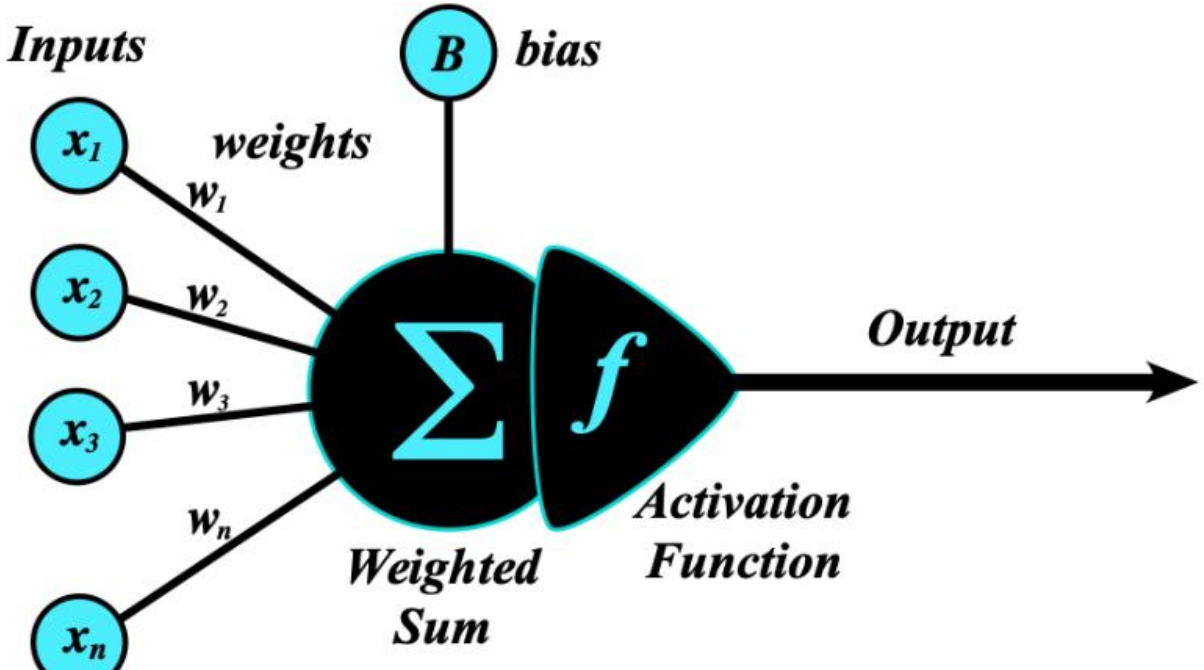
Strongly disagree

Strongly agree

ANN – Neural Networks

- **Idee:** Replicarea mecanismelor din creierul uman

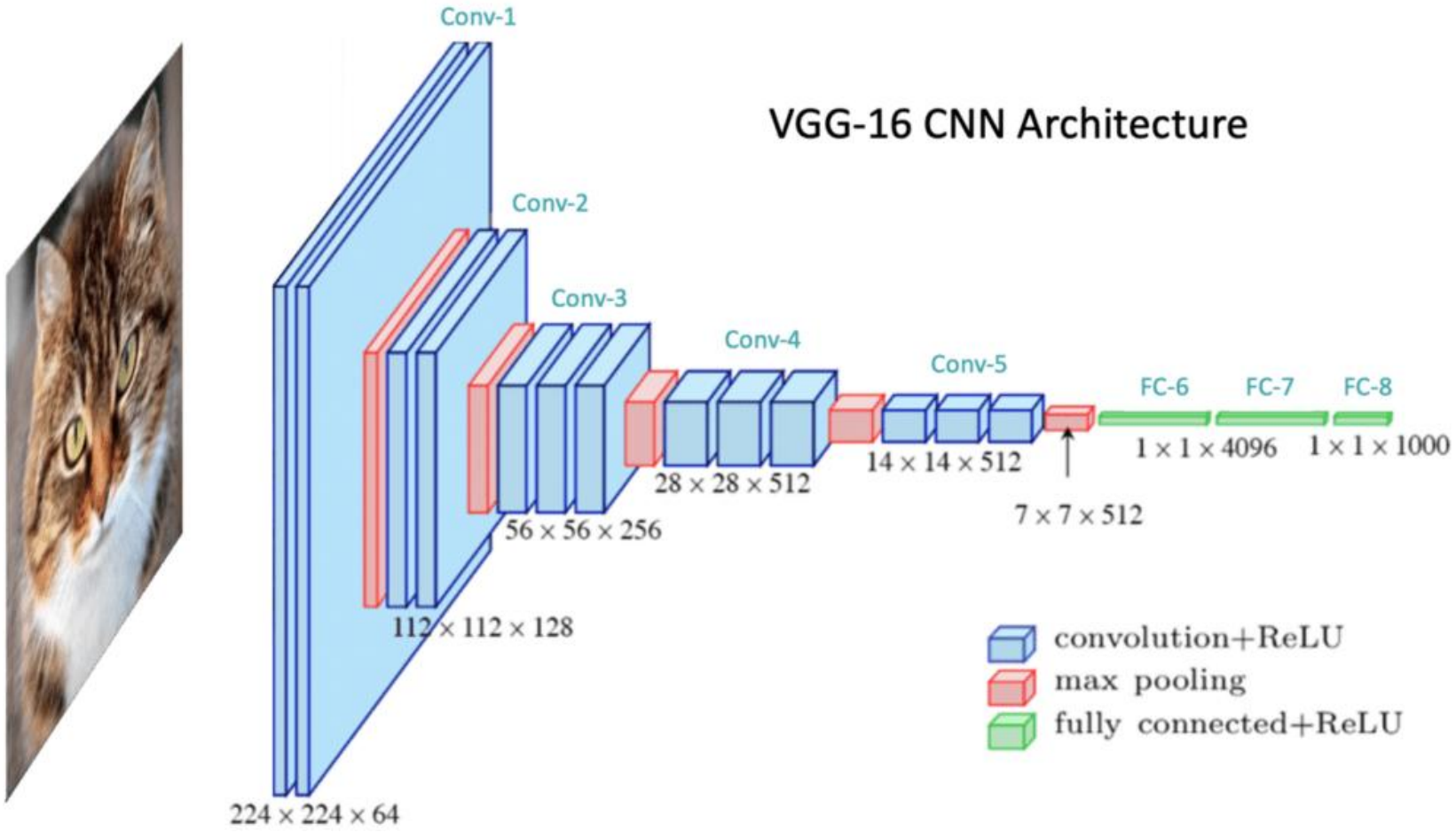

Dendrites
Axon
Cell body
Terminals


Inputs
B
bias
weights
x_1
w_1
x_2
w_2
x_3
w_3
x_n
w_n
Σ
f
Output
Activation Function
Weighted Sum

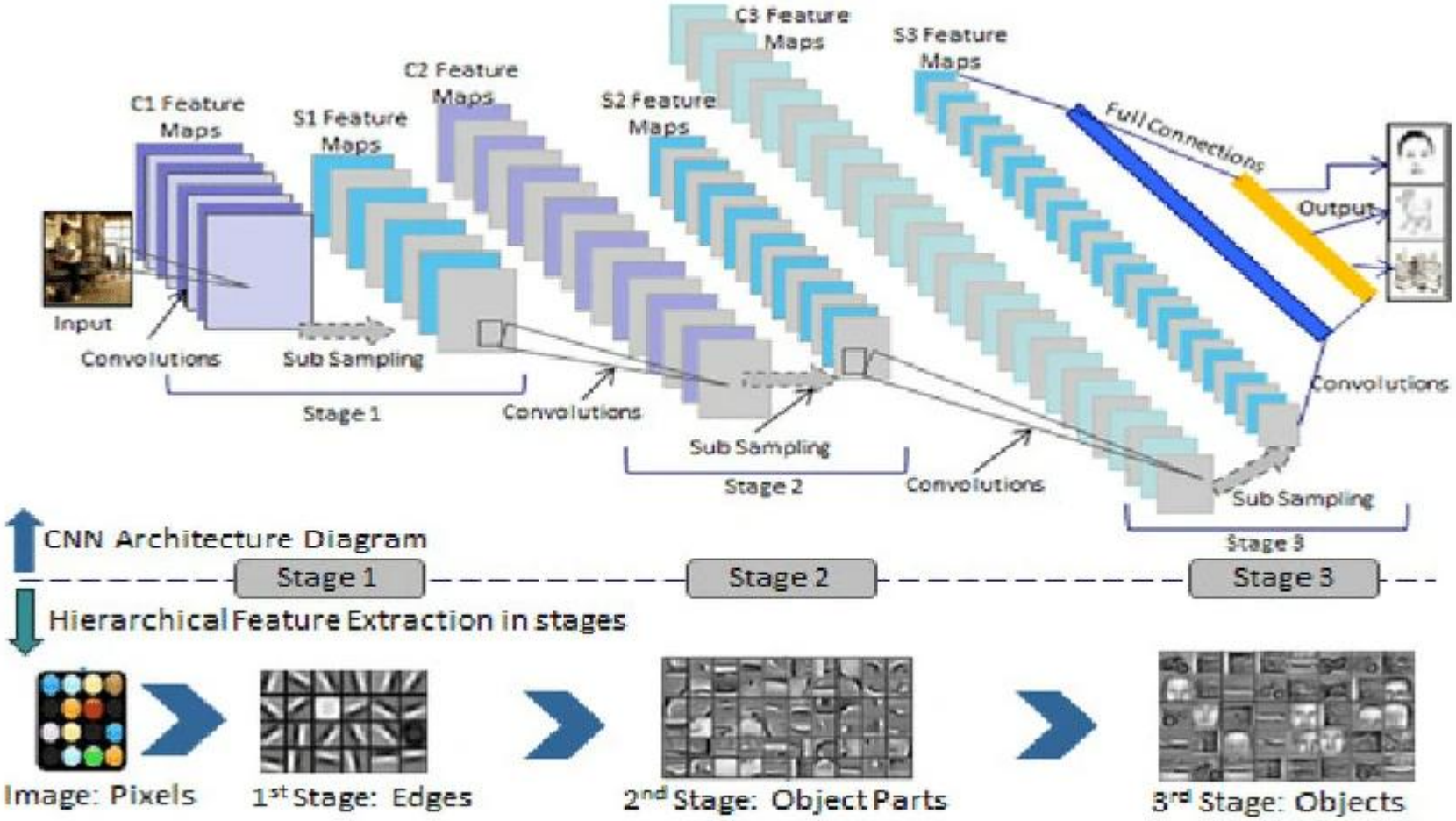
CNN – Convolutional Neural Networks

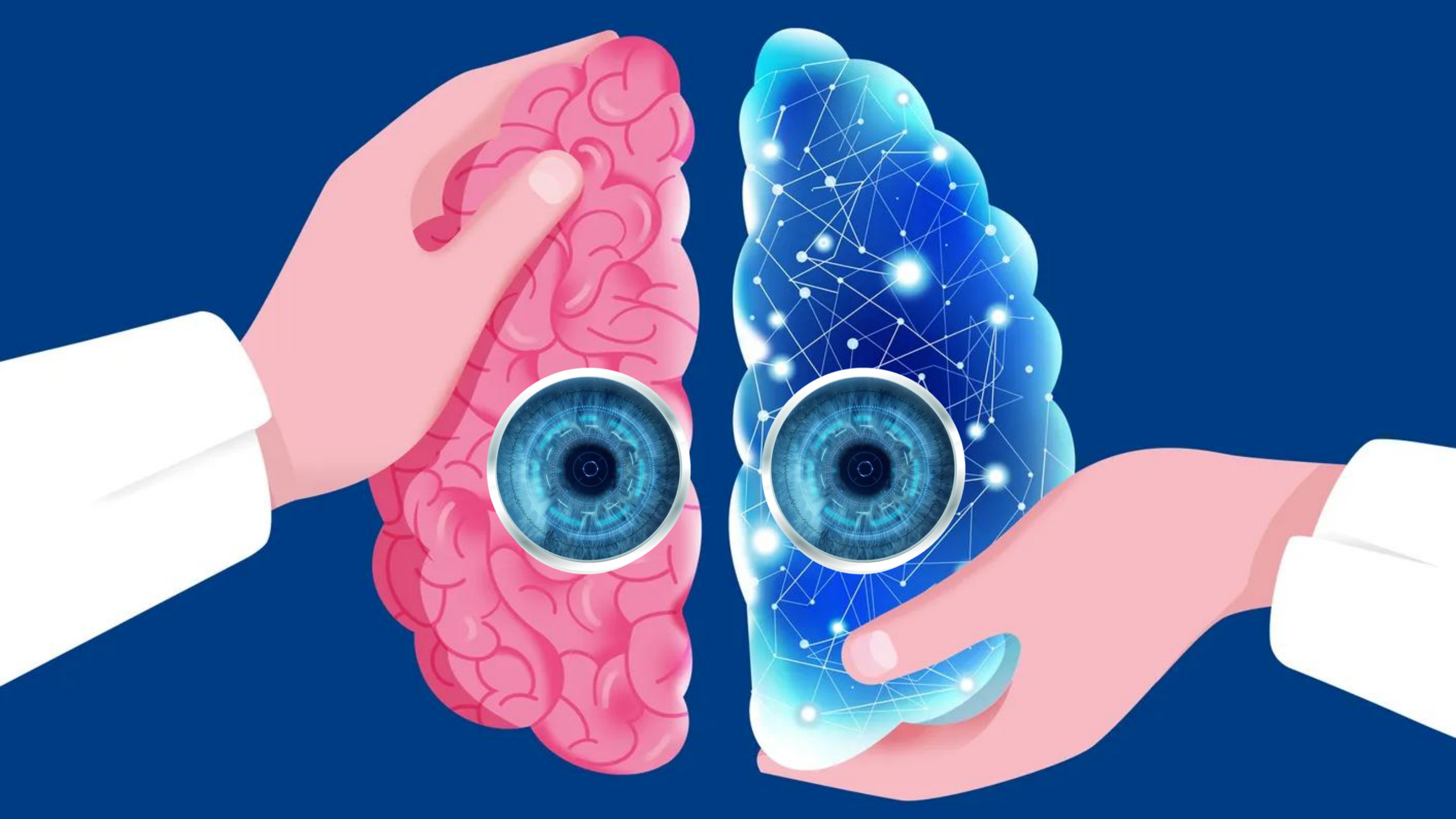
- **Idee:** Replicarea mecanismului din cortexul visual
- Introduse și aplicate cu succes în Computer Vision
- **Vizualizare:** https://hannibunny.github.io/mlbook/neuralnetworks/convolutionDemos.html

CNN – Convolutional Neural Networks

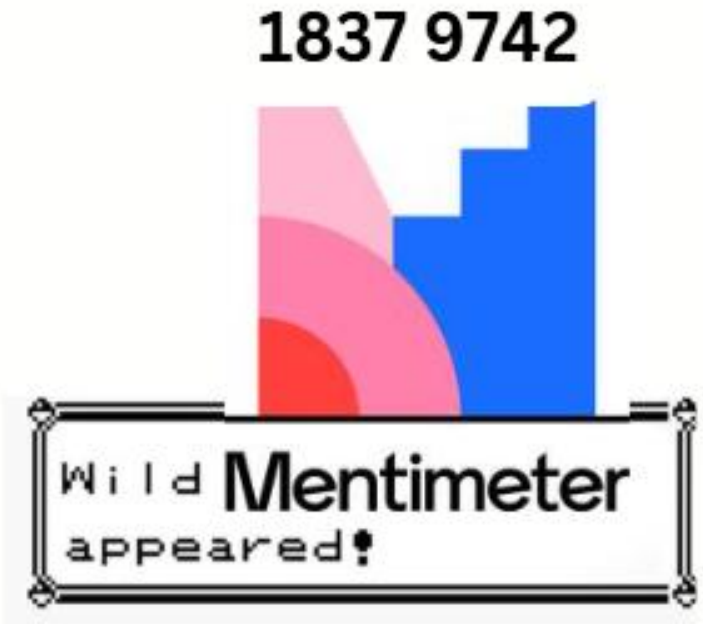


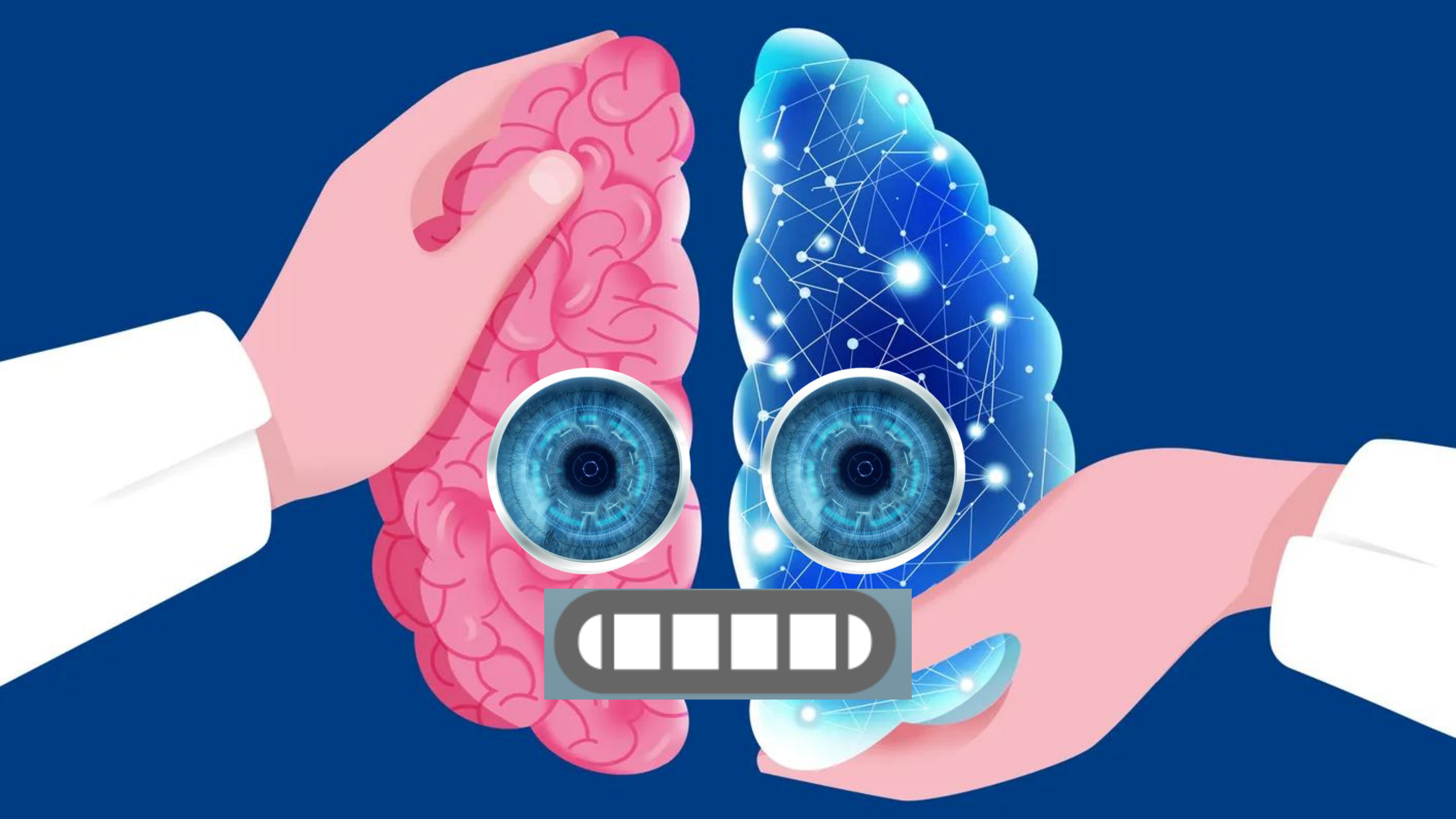
CNN – Convolutional Neural Networks





Ce lipsește?





Natural Language Processing?

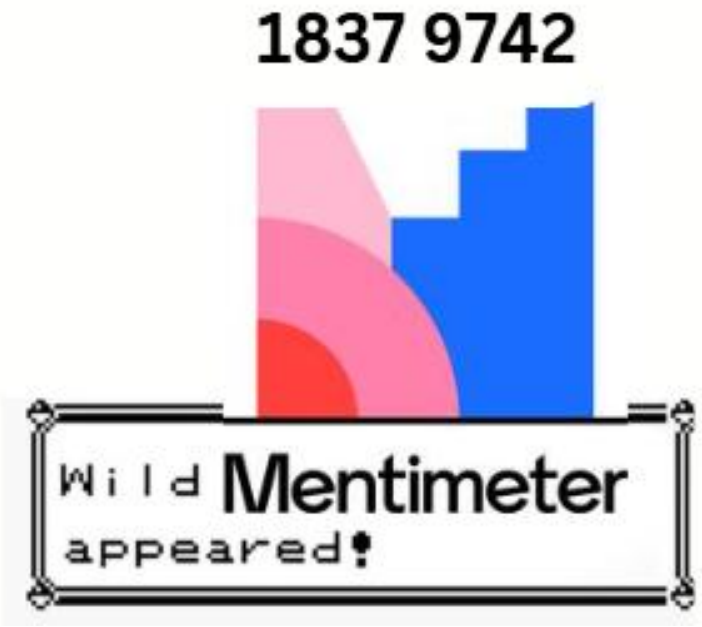
NLP Tasks
Text Preprocessing
Coreference Resolution
Part Of Speech Tagging
Word Sense Disambiguation
Grammatical Error Correction
Feature Extraction
Text-to-Data and viceversa
Speech-to-Text
Text-to-Image
Data-to-Text
Text-to-Speech
Fake News and Hate Speech Detection
Fake News Detection
Stance Detection
Hate Speech Detection
Text Reasoning
Common Sense Reasoning
Natural Language Inference
Classification
Text Classification
Sentiment Analysis
Chatbots
Slot Filling
Intent Detection
Keyword Extraction
Information Retrieval and Document Ranking
Sentence/Document Similarity
Question Answering
Topics and Keywords
Keyword Extraction
Topic Modeling
Text-to-Text Generation
Machine Translation
Text Generation
Text Summarization
Text Simplification
Lexical Normalization
Paraphrase Generation
Knowledge Bases, Entities and Relations
Relation Extraction
Relation Prediciton
Named Entity Recognition
Entity Linking
Image by NLPlanet
NLP

NLP vs CV

Imagine: Date Dense, Continue
Text: Date Discrete, Ambigue

CNN: capturează dependențe locale
Text: Este nevoie de context larg

Ce concept a fost introdus (~2014-2017)?

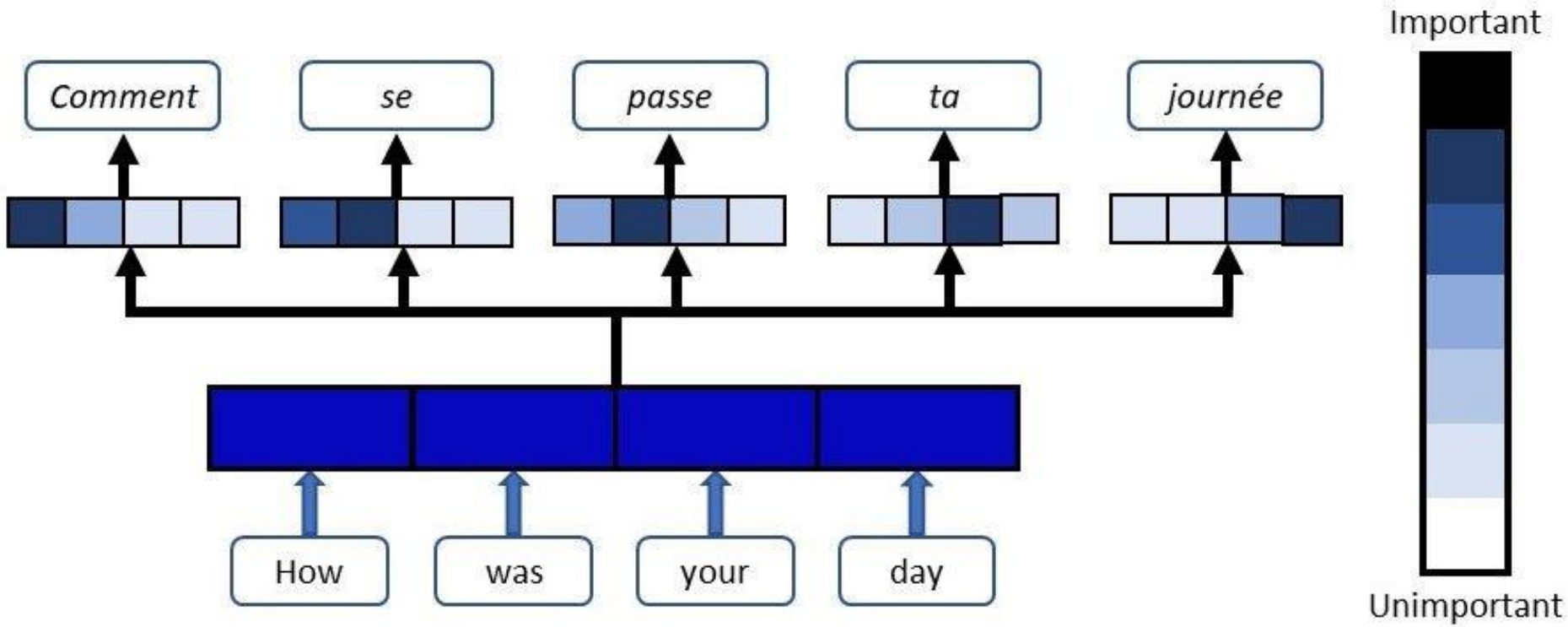


Attention is all you need

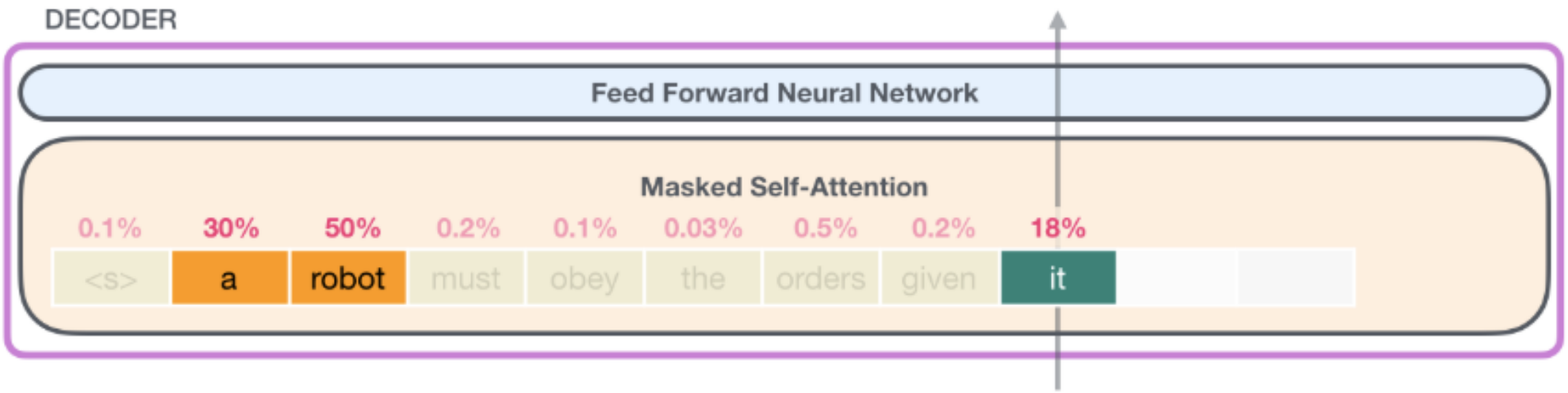
- Introdus pentru “Machine Translation” [1]
- **Concept:** Învățăm să ne “concentrăm” pe zone de interes, relații între anumite secvențe din input
- Asignăm ponderi (weights) la diferite părți din input

[1] Bahdanau, D., Cho, K. and Bengio, Y., 2014. Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473.

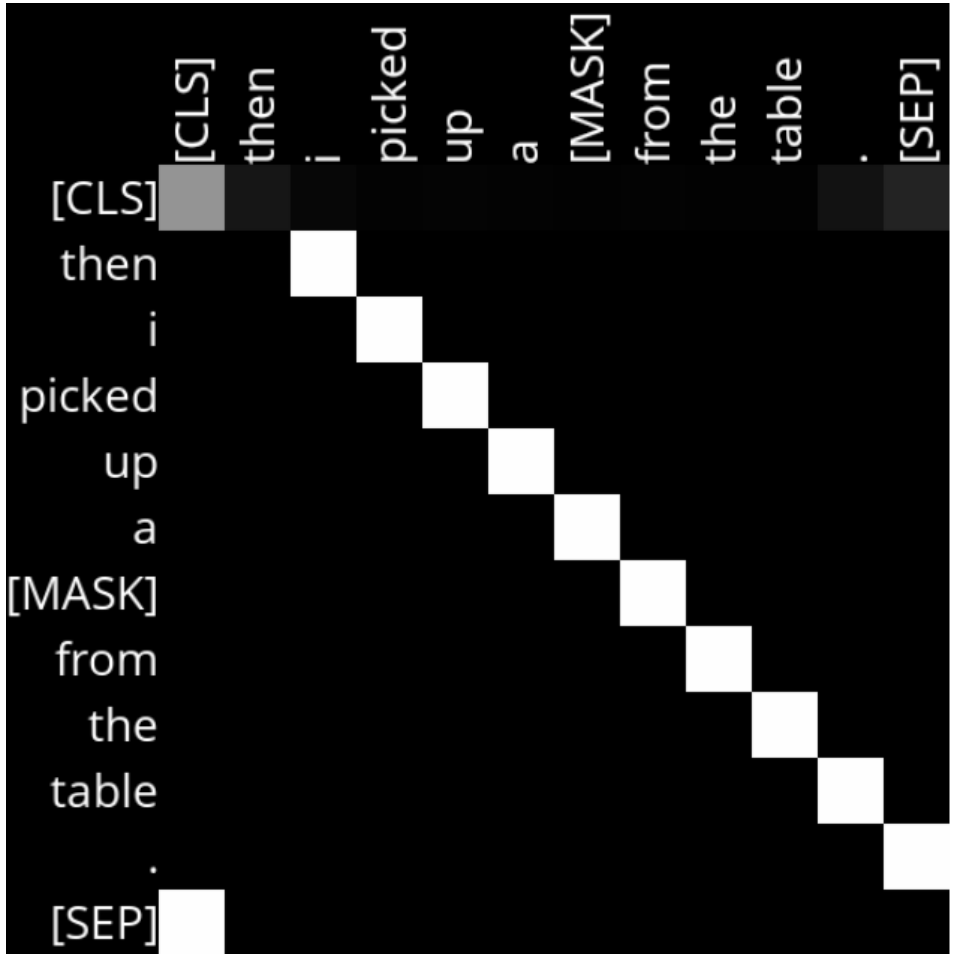
Attention

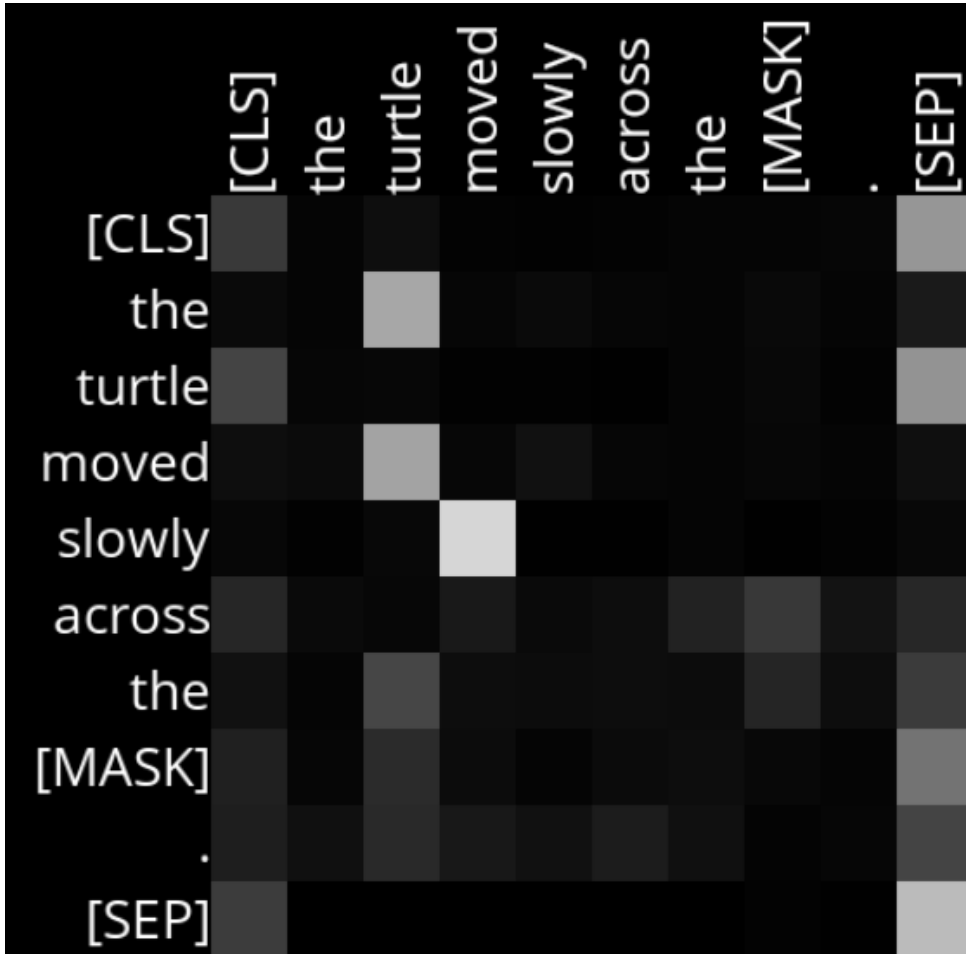

Comment
se
passe
ta
journée
How
was
your
day
Important
Unimportant

Attention


DECODER
Feed Forward Neural Network
Masked Self-Attention
0.1%
30%
50%
0.2%
0.1%
0.03%
0.5%
0.2%
18%
<s>
a
robot
must
obey
the
orders
given
it

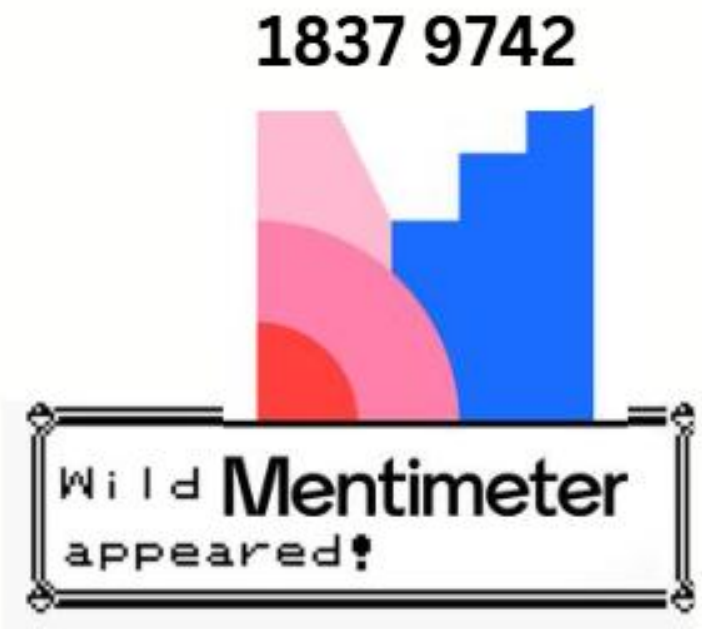
Attention



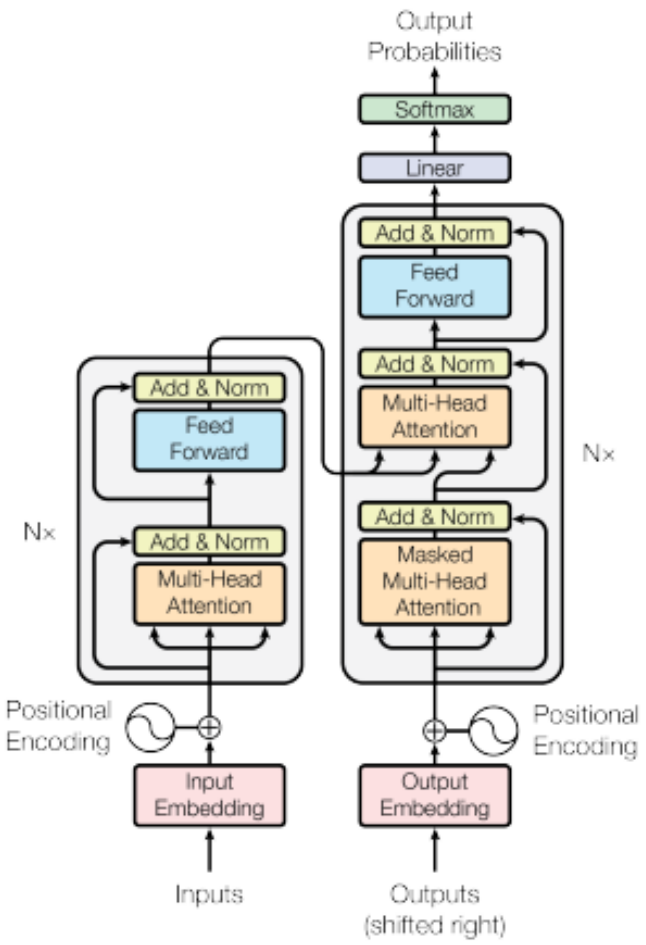


https://cs50.harvard.edu/ai/projects/6/attention/

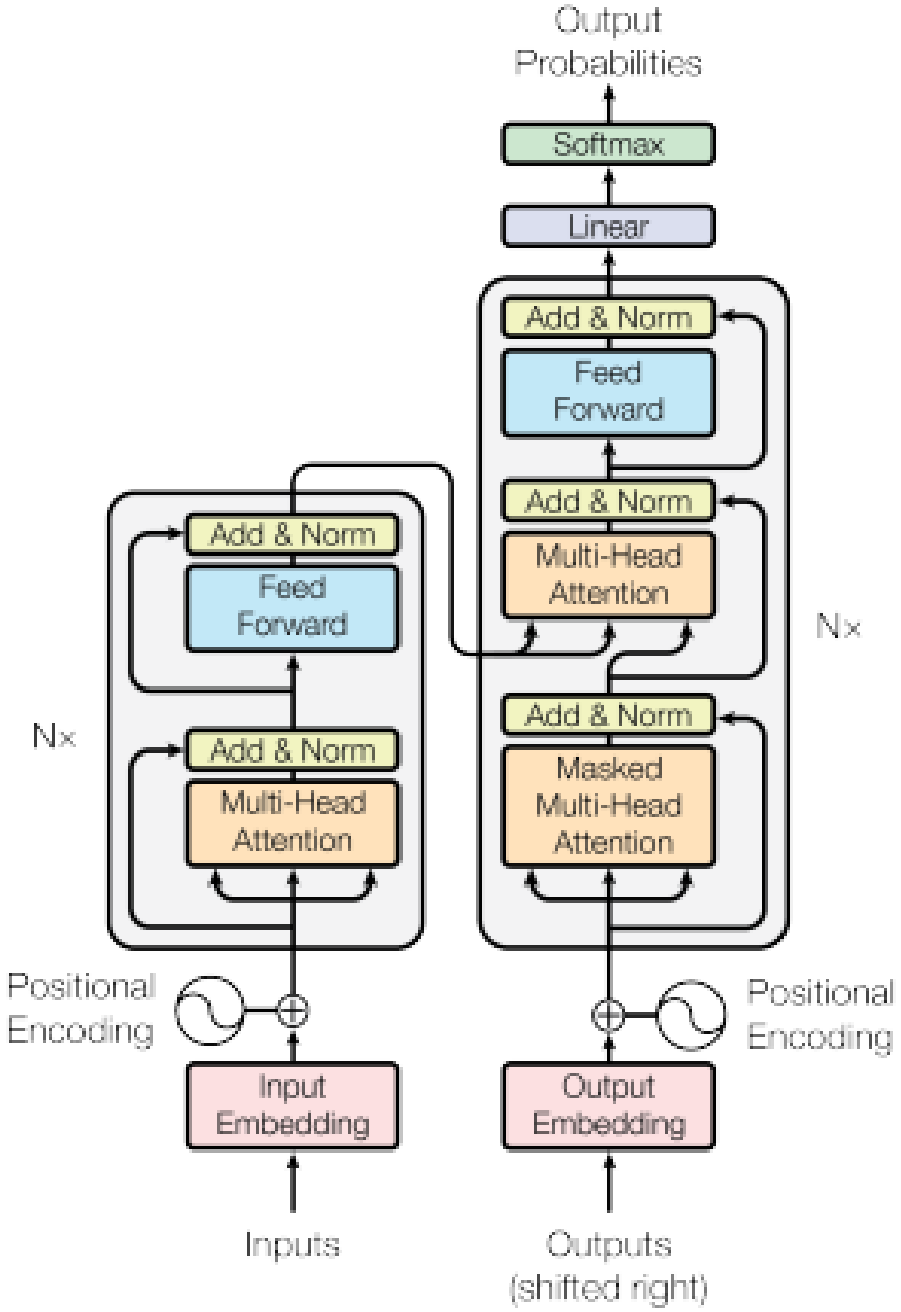
Attention “focus” check



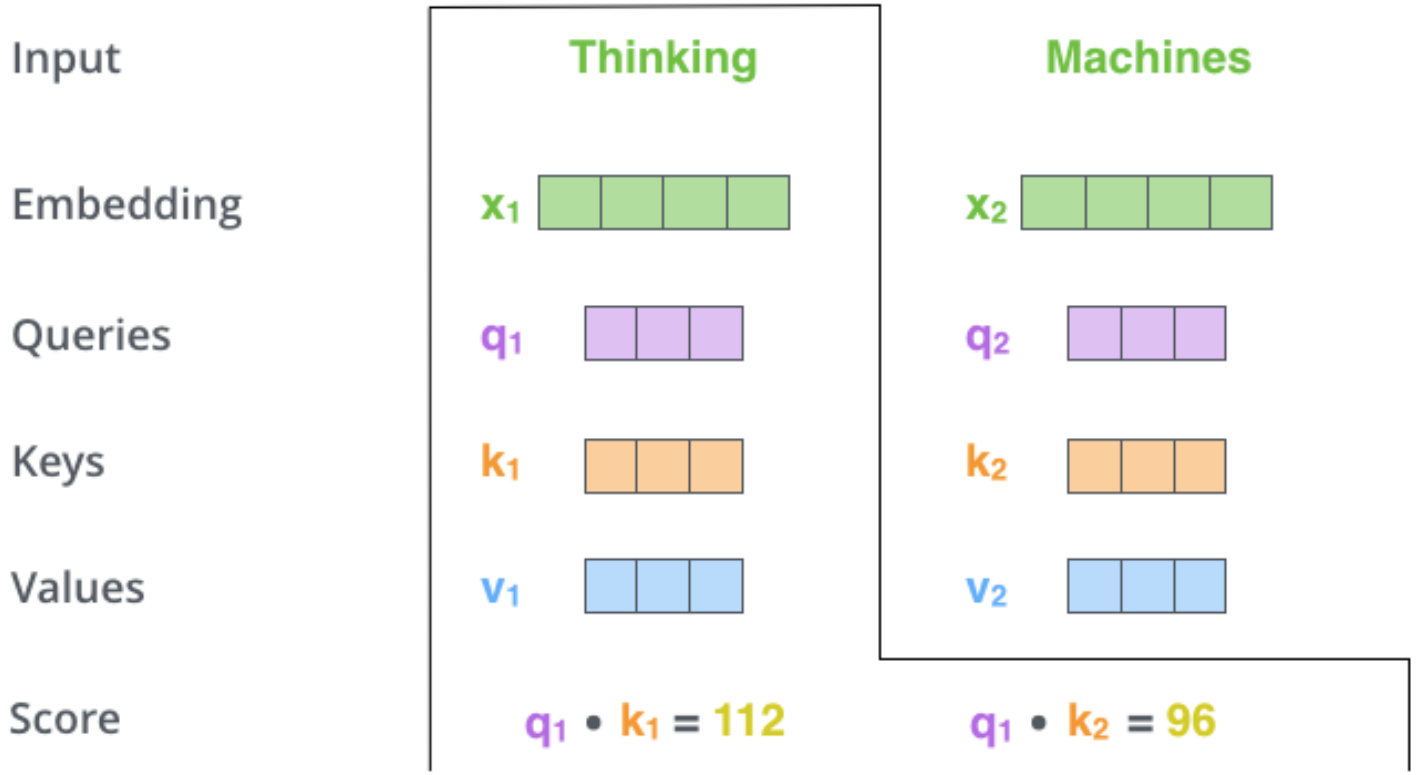
Transformers [2]



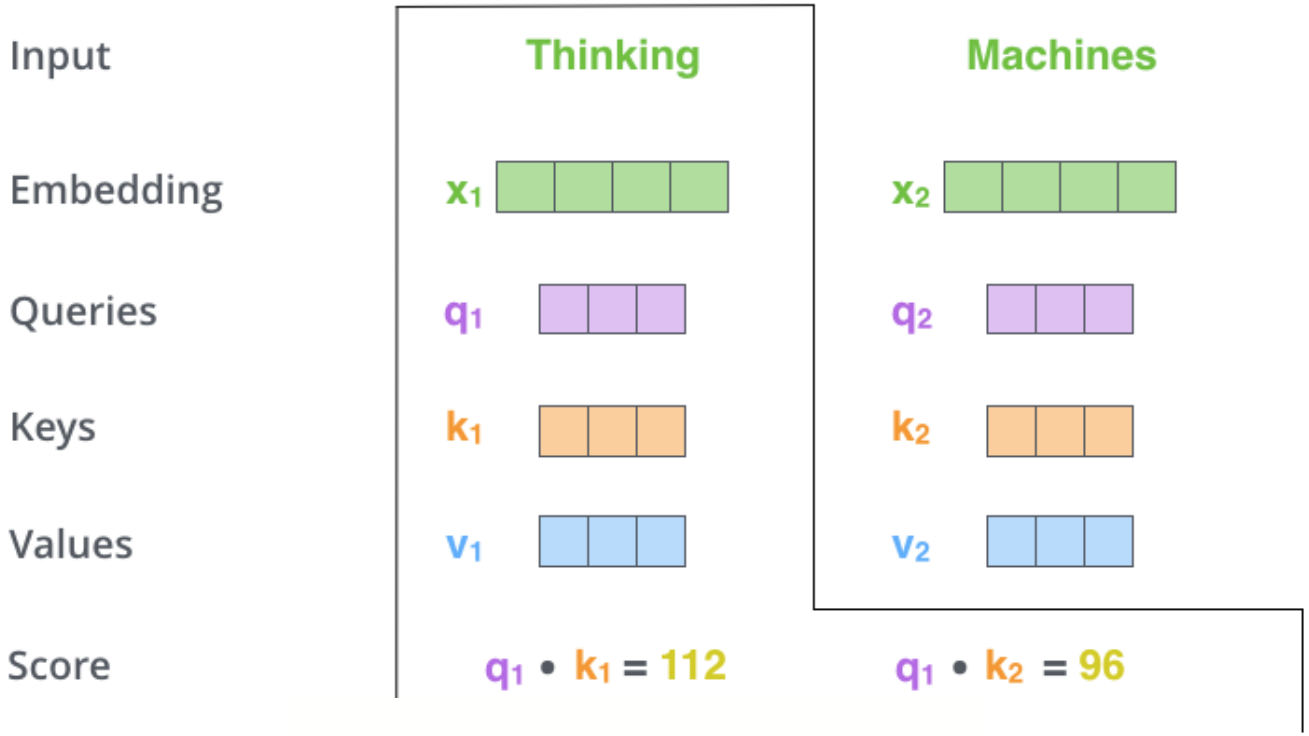
[2] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. Attention is all you need. *Advances in neural information processing systems*, *30*.


Output
Probabilities
Softmax
Linear
Add & Norm
Feed
Forward
Add & Norm
Multi-Head
Attention
Add & Norm
Masked
Multi-Head
Attention
Nx
Add & Norm
Feed
Forward
Add & Norm
Multi-Head
Attention
Nx
Positional
Encoding
Positional
Encoding
Input
Embedding
Output
Embedding
Inputs
Outputs
(shifted right)

Attention as Key, Query, Value



[2] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. Attention is all you need. *Advances in neural information processing systems*, *30*.



1837 9742

Wild Mentimeter appeared!

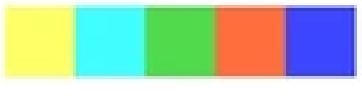
Introducere în Generative AI

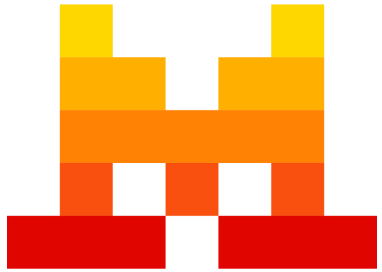
ChatGPT

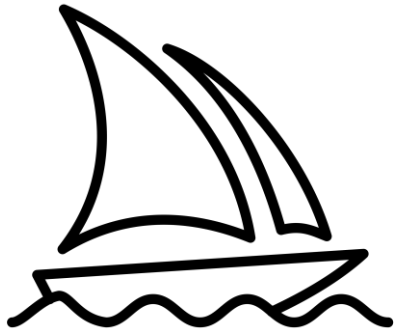
Gemini

Claude

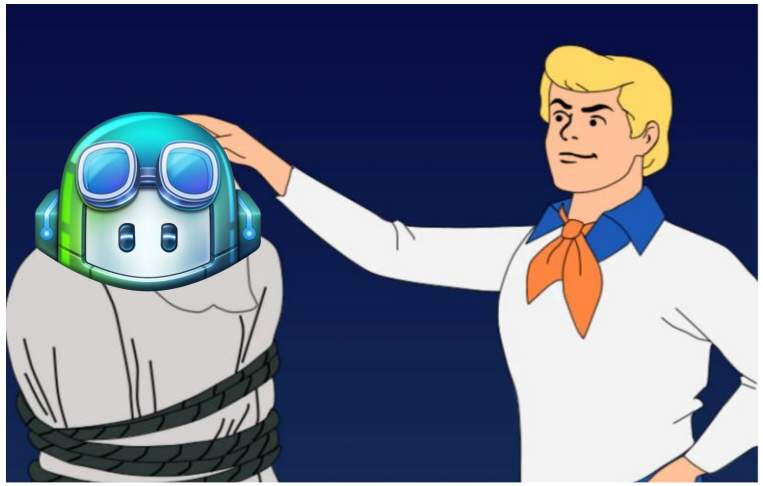
DALL·E





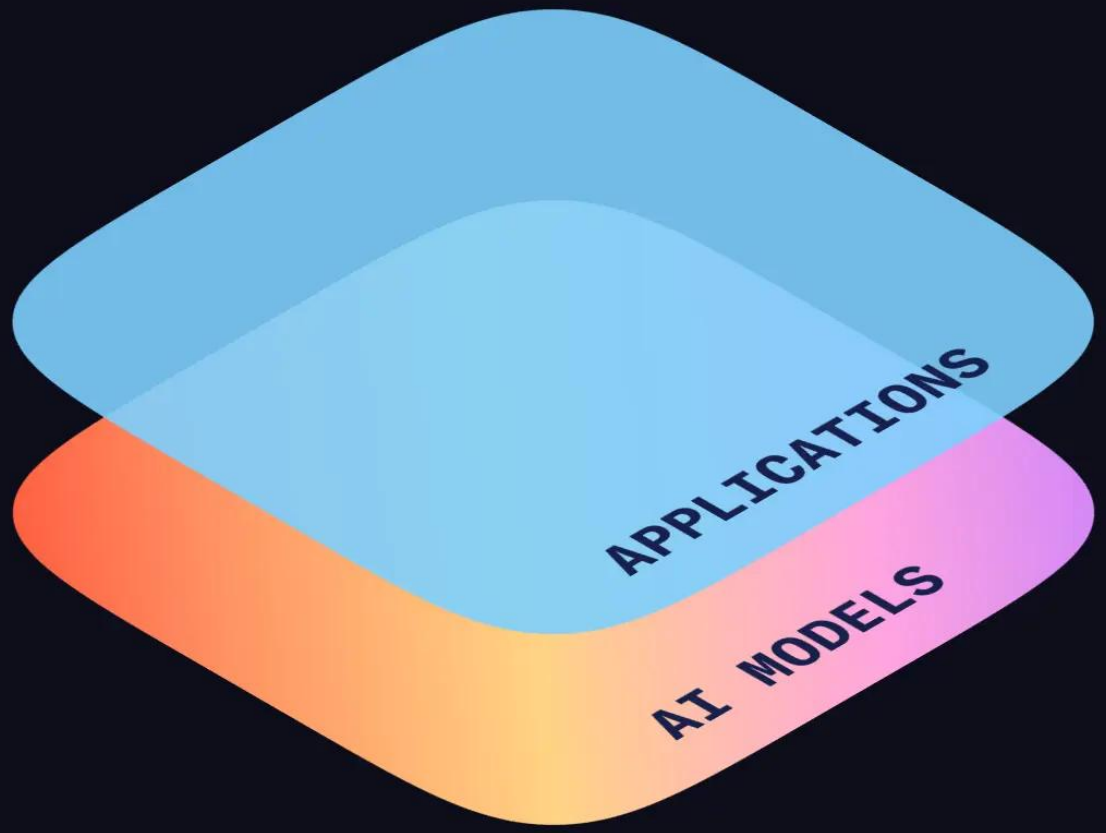


Introducere în Generative AI



- ChatGPT, Gemini, Copilot, Claude sunt sisteme software / tool-uri complexe
- Aceste au în spate modele inteligente generative (LLMs)

The Generative Tech Stack



TOOLS THAT COLLABORATE WITH AI MODELS.

WORKFLOWS, SECURITY, NETWORK EFFECTS, PAYMENTS, ETC. (10,000'S OF THESE)

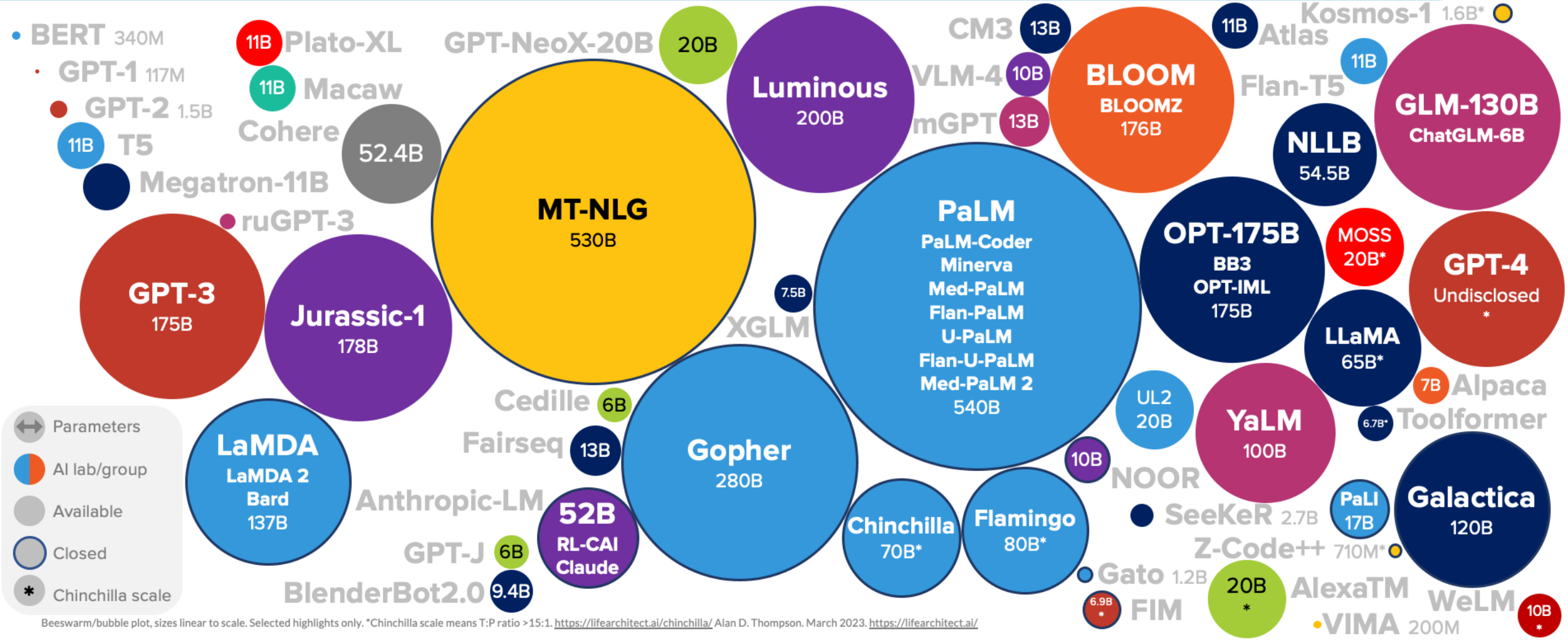
MODELS GENERATE UNIQUE AND NOVEL OUTPUT.

GPT-3, STABLE DIFFUSION, CUSTOM DATA SETS, ETC. (1,000'S OF THESE)

SOURCE: NFX.COM (2022)

NfX

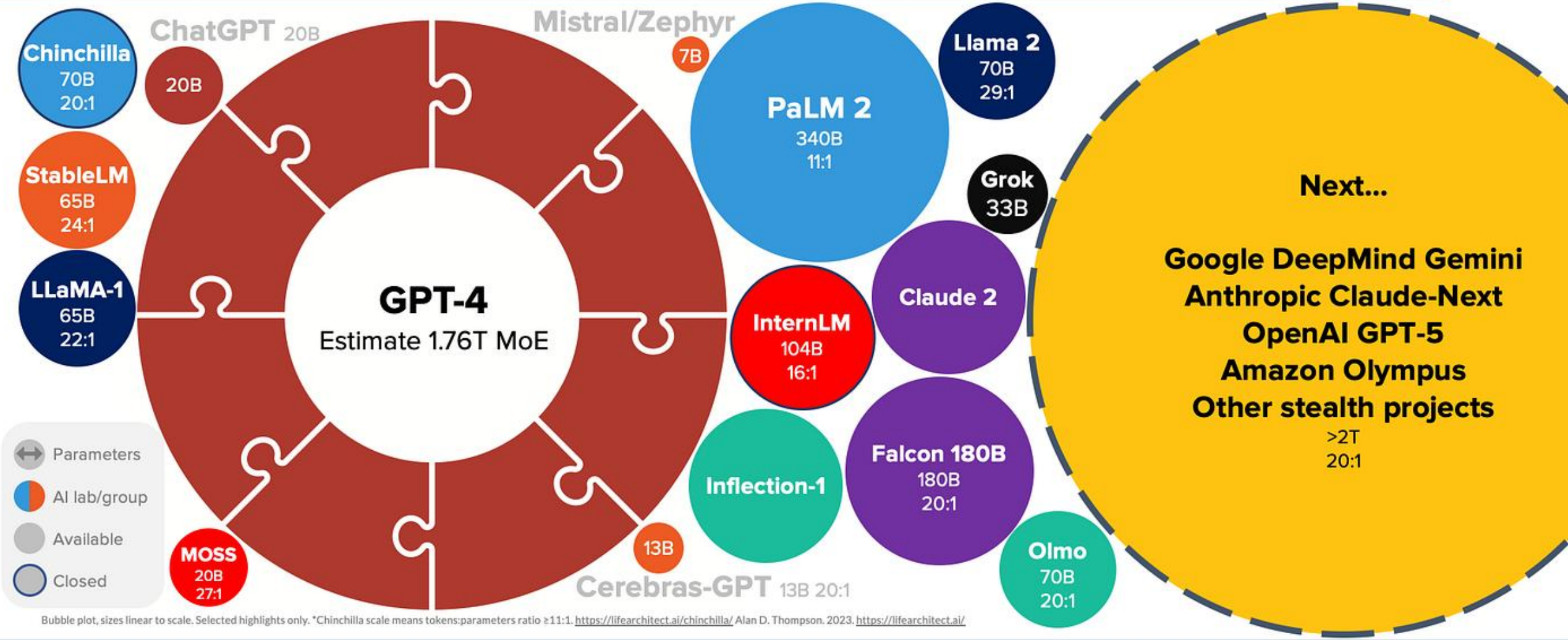
LANGUAGE MODEL SIZES TO MAR/2023



Beeswarm/bubble plot, sizes linear to scale. Selected highlights only. *Chinchilla scale means T:P ratio >15:1. https://lifearchitect.ai/chinchilla/ Alan D. Thompson. March 2023. https://lifearchitect.ai/

2023-2024 OPTIMAL LANGUAGE MODELS

NOV/ 2023



Bubble plot, sizes linear to scale. Selected highlights only. *Chinchilla scale means tokens:parameters ratio ≥11:1. https://lifearchitect.ai/chinchilla/ Alan D. Thompson. 2023. https://lifearchitect.ai/

LifeArchitect.ai/models

Model	Lab	Playground	Parameters (B)	Tokens trained (B)	Ratio Tokens:Params (Chinchilla scaling≥20:1)	ALScore "ALScore" i Sqr Root of	MMLU	MMLU -Pro	GPQA	HLE	Training dataset	Announced ▼	Public?	Paper / Repo	Arch	Tags
AuroraGPT (ScienceGF	Argonne National L	https://lifearchite	*2000*	*30000*	15:1							*TBA*				
DeepSeek-R2	DeepSeek-AI	https://www.reut										*TBA*			*MoE*	Reasoning SOTA
ERNIE 5	Baidu	https://lifearchite										*TBA*				
Grok-4	xAI	https://lifearchite										*TBA*			*MoE*	Reasoning SOTA
GPT-5	OpenAI	https://lifearchite	*5400*	*114000*	22:1							*TBA*			*MoE*	SOTA
GPT-6	OpenAI	https://lifearchite										*TBA*				SOTA
Llama 4 Reasoning	Meta AI	https://ai.meta.cc										*TBA*		https://ai.m	MoE	SOTA Reasoning
~~*MAI-1*~~	~~Microsoft~~	~~https://arstechni~~	~~500~~	~~*10000*~~	~~20:1~~	~~7.5~~						~~*TBA*~~		~~https://www~~	~~*Dense*~~	
o4	OpenAI	https://lifearchite										*TBA*				Reasoning SOTA
Llama 4 Behemoth	Meta AI	https://ai.meta.cc	2000	30000	15:1	25.8		82.2	73.7			Apr/2025		https://ai.m	MoE	SOTA
Llama 4 Maverick	Meta AI	https://ai.meta.cc	400	22000	55:1	9.9		80.5	69.8			Apr/2025		https://ai.m	MoE	SOTA
Qwerky-72B	Featherless AI	https://featherles	72	18000	250:1	3.8	77.46				synthetic, web-scale	Apr/2025		https://featl	Dense	
Agentic-Tx	Google DeepMind	https://github.co	*200*	*20000*	100:1	6.7			62.4	14.5	synthetic, web-scale	Mar/2025		https://stor	*MoE*	
TxGemma	Google DeepMind	https://huggingfa	27	14000	519:1	2.0					synthetic, web-scale	Mar/2025		https://stor	*MoE*	Reasoning
Gemini 2.5 Pro	Google DeepMind	https://aistudio.g	*200*	*20000*	100:1	6.7			84	18.8	synthetic, web-scale	Mar/2025		https://blog	*MoE*	Reasoning SOTA
DeepSeek-V3 0324	DeepSeek-AI	https://chat.deep	685	14800	22:1	10.6		81.2	68.4		synthetic, web-scale	Mar/2025		https://hugg	MoE	SOTA
Llama-3.3-Nemotron-	NVIDIA	https://huggingfa	49	*15040*	307:1	2.9			66.67		web-scale	Mar/2025		https://buil	Dense	Reasoning
EXAONE Deep	LG	https://huggingfa	32	6500	204:1	1.5	83	74	66.1		web-scale	Mar/2025		https://arxi	Dense	Reasoning
Mistral Small 3.1	Mistral	https://huggingfa	24	*8000*	334:1	1.5	81.01	56.03	37.5		web-scale	Mar/2025		https://mist	Dense	
ERNIE 4.5	Baidu	https://yiyan.bai						*79*	*55*		synthetic, web-scale	Mar/2025		https://www	*MoE*	
X1	Baidu	https://yiyan.bai									synthetic, web-scale	Mar/2025		https://www	*MoE*	Reasoning
OLMo 2 32B	Allen AI	https://playgrour	32	6400	200:1	1.5	78				synthetic, web-scale	Mar/2025		https://alle	Dense	
Command A	Cohere	https://dashboar	111	*8000*	73:1	3.1	85				synthetic, web-scale	Mar/2025		https://hugg	Dense	
Gemini Robotics	Google DeepMind		*200*	*20000*	100:1	6.7		*79.1*	*64.7*		synthetic, web-scale	Mar/2025		https://stor	*MoE*	

https://lifearchitect.ai/ - 2025

Join at menti.com | use code 1837 9742

Mentimeter

Instructions

Go to

www.menti.com

Enter the code

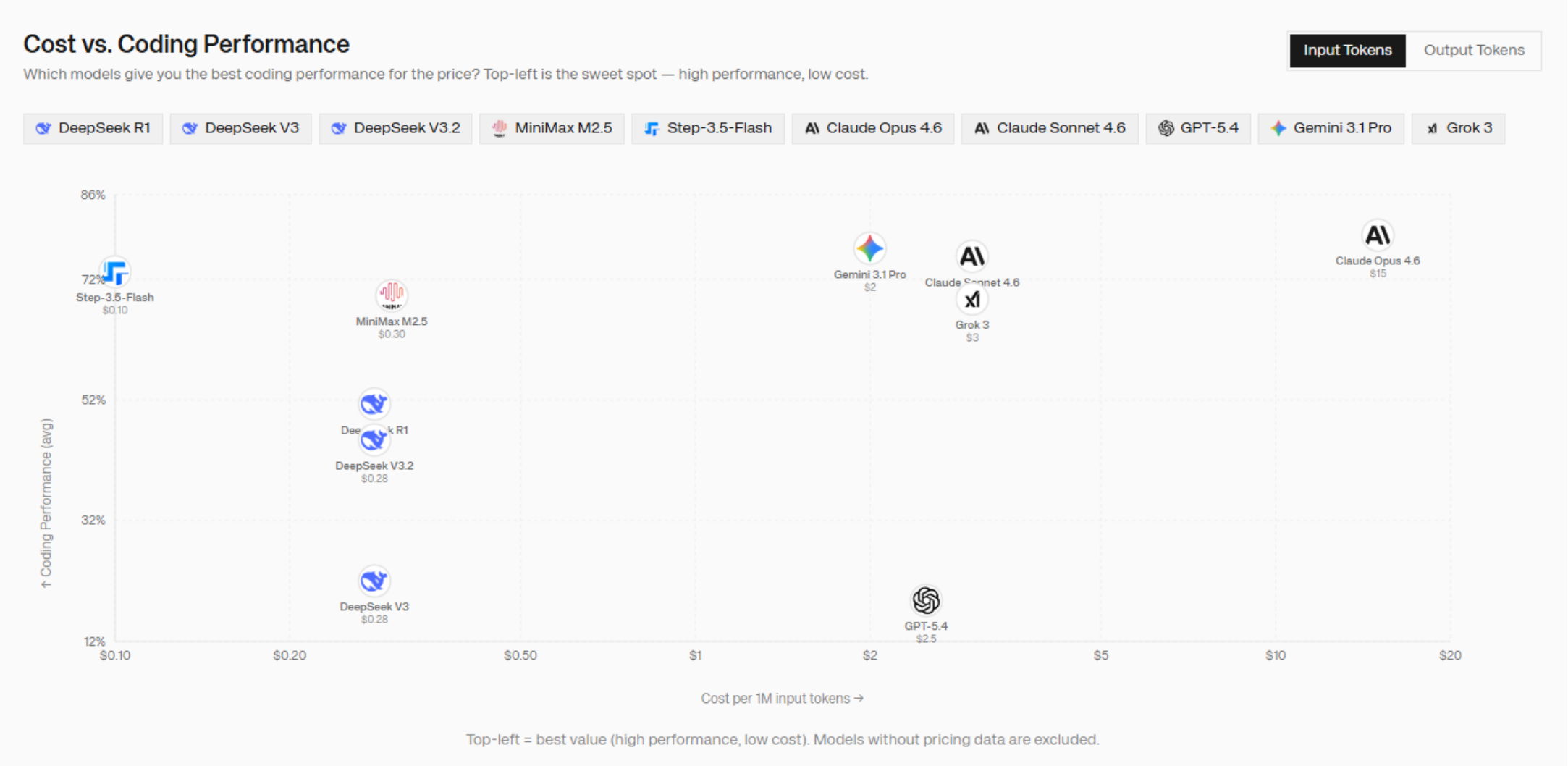
1837 9742

Or use QR code

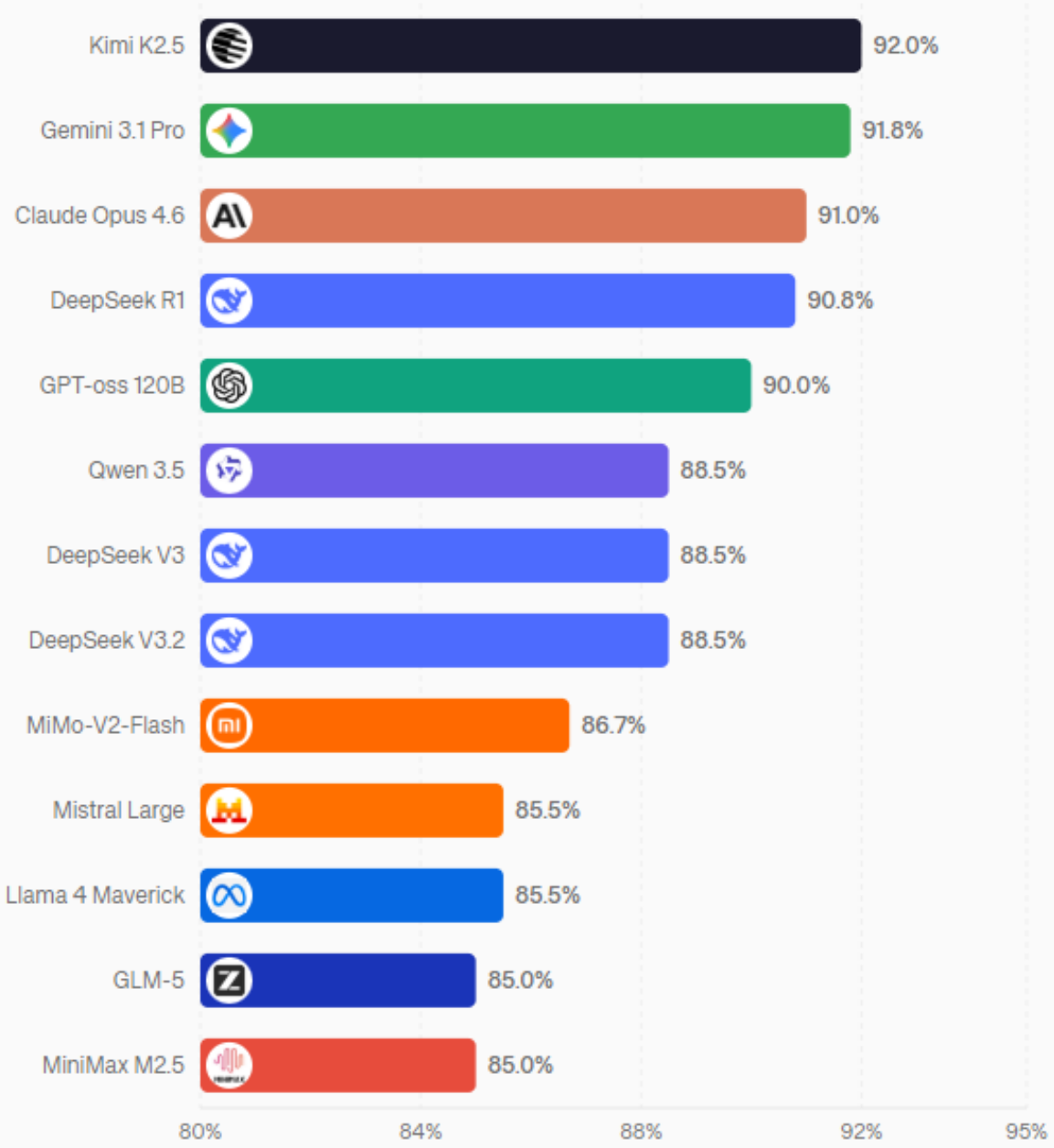
LLM Leaderboards

- https://onyx.app/llm-leaderboard
- https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard#/
- https://huggingface.co/spaces/ArtificialAnalysis/LLM-Performance-Leaderboard
- https://arena.ai/leaderboard/

Onyx Leaderboard



Best Overall (MMLU)



Open LLM Leaderboard (HuggingFace)

Rank	Type	Model	Average	IFEval	BBH	MATH	GPQA	MUSR	MMLU-PRO	CO_2 Cost
1		MaziyarPanahi/calme-3.2-instruct-78b	52.08 %	80.63 %	62.61 %	40.33 %	20.36 %	38.53 %	70.03 %	66.01 kg
2		MaziyarPanahi/calme-3.1-instruct-78b	51.29 %	81.36 %	62.41 %	39.27 %	19.46 %	36.50 %	68.72 %	64.44 kg
3		dfurman/CalmeRys-78B-Orpo-v0.1	51.23 %	81.63 %	61.92 %	40.63 %	20.02 %	36.37 %	66.80 %	25.99 kg
4		MaziyarPanahi/calme-2.4-rys-78b	50.77 %	80.11 %	62.16 %	40.71 %	20.36 %	34.57 %	66.69 %	25.95 kg
5		huihui-ai/Qwen2.5-72B-Instruct-abliterated	48.11 %	85.93 %	60.49 %	60.12 %	19.35 %	12.34 %	50.41 %	76.77 kg
6		Qwen/Qwen2.5-72B-Instruct	47.98 %	86.38 %	61.87 %	59.82 %	16.67 %	11.74 %	51.40 %	47.65 kg
7		MaziyarPanahi/calme-2.1-qwen2.5-72b	47.86 %	86.62 %	61.66 %	59.14 %	15.10 %	13.30 %	51.32 %	29.50 kg
8		newsbang/Homer-v1.0-Qwen2.5-72B	47.46 %	76.28 %	62.27 %	49.02 %	22.15 %	17.90 %	57.17 %	29.55 kg
9		ehristoforu/qwen2.5-test-32b-it	47.37 %	78.89 %	58.28 %	59.74 %	15.21 %	19.13 %	52.95 %	29.54 kg
10		Saxo/Linkbricks-Horizon-AI-Avengers-V1-32B	47.34 %	79.72 %	57.63 %	60.27 %	14.99 %	18.16 %	53.25 %	7.95 kg

Hugging Face Leaderboard API

API Provider	Model	Context Window	License	Artificial Analysis Intelligence Index	Blended USD/1M Tokens	Median Tokens/s	Median First Chunk (s)	Total Response (s)	Reasoning Time (s)	Further Analysis
Google (AI Studio)	Gemini 3.1 Pro Preview (AI Studio)	1m	Proprietary	57	$4.50	111	29.71	34.23	N/A	Model Pr
Google (Vertex)	Gemini 3.1 Pro Preview (Vertex)	1m	Proprietary	57	$4.50	108	36.47	41.11	N/A	Model Pr
OpenAI	GPT-5.4 (xhigh)	1m	Proprietary	57	$5.63	74	151.79	158.58	N/A	Model Pr
Microsoft Azure	GPT-5.4 (xhigh)	1m	Proprietary	57	$5.63	56	195.06	204.03	N/A	Model Pr
OpenAI	GPT-5.3 Codex (xhigh)	400k	Proprietary	54	$4.81	79	83.90	90.26	N/A	Model Pr
Anthropic	Claude Opus 4.6 (max)	1m	Proprietary	53	$10.00	42	7.97	20.01	N/A	Model Pr
Microsoft Azure	Claude Opus 4.6 (max)	1m	Proprietary	53	$10.00	41	8.79	20.93	N/A	Model Pr

Arena AI Leaderboard

Leaderboard Overview

See how leading AI models stack up across text, image, vision, and more. This page provides a high-level snapshot of each Arena. Explore dedicated tabs for deeper insights. Learn more here.

Text

4 days ago

Rank	Model	Score	Votes
1	claude-opus-4-6-thinking	1504	12,730
2	claude-opus-4-6	1500	13,553
3	gemini-3.1-pro-preview	1493	15,809
4	grok-4.20-beta1	1491	7,378
5	gemini-3-pro	1486	41,631
6	gpt-5.4-high	1484	5,570
7	grok-4.20-beta-0309-reaso...	1483	5,702
8	gpt-5.2-chat-latest-20260...	1480	11,405
9	gemini-3-flash	1474	30,962
10	claude-opus-4-5-20251101-...	1474	37,448

View all

Code

4 days ago

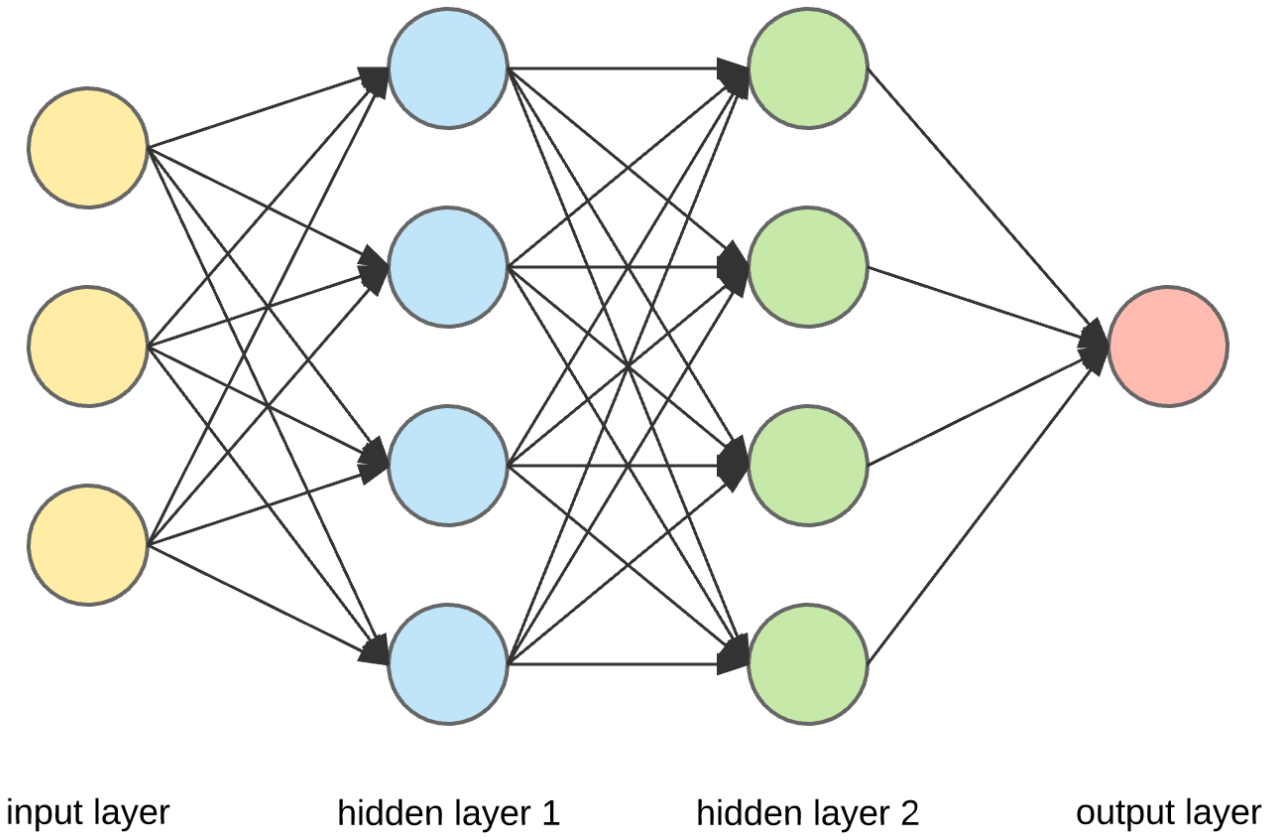
Rank	Model	Score	Votes
1	claude-opus-4-6	1549	4,264
2	claude-opus-4-6-thinking	1545	3,495
3	claude-sonnet-4-6	1523	6,391
4	claude-opus-4-5-20251101-...	1491	13,247
5	claude-opus-4-5-20251101	1465	13,559
6	gpt-5.4-high (codex-harne...	1457	1,488
7	gemini-3.1-pro-preview	1455	4,733
8	glm-5	1445	4,265
9	glm-4.7	1439	4,877
10	gemini-3-pro	1438	17,152

View all

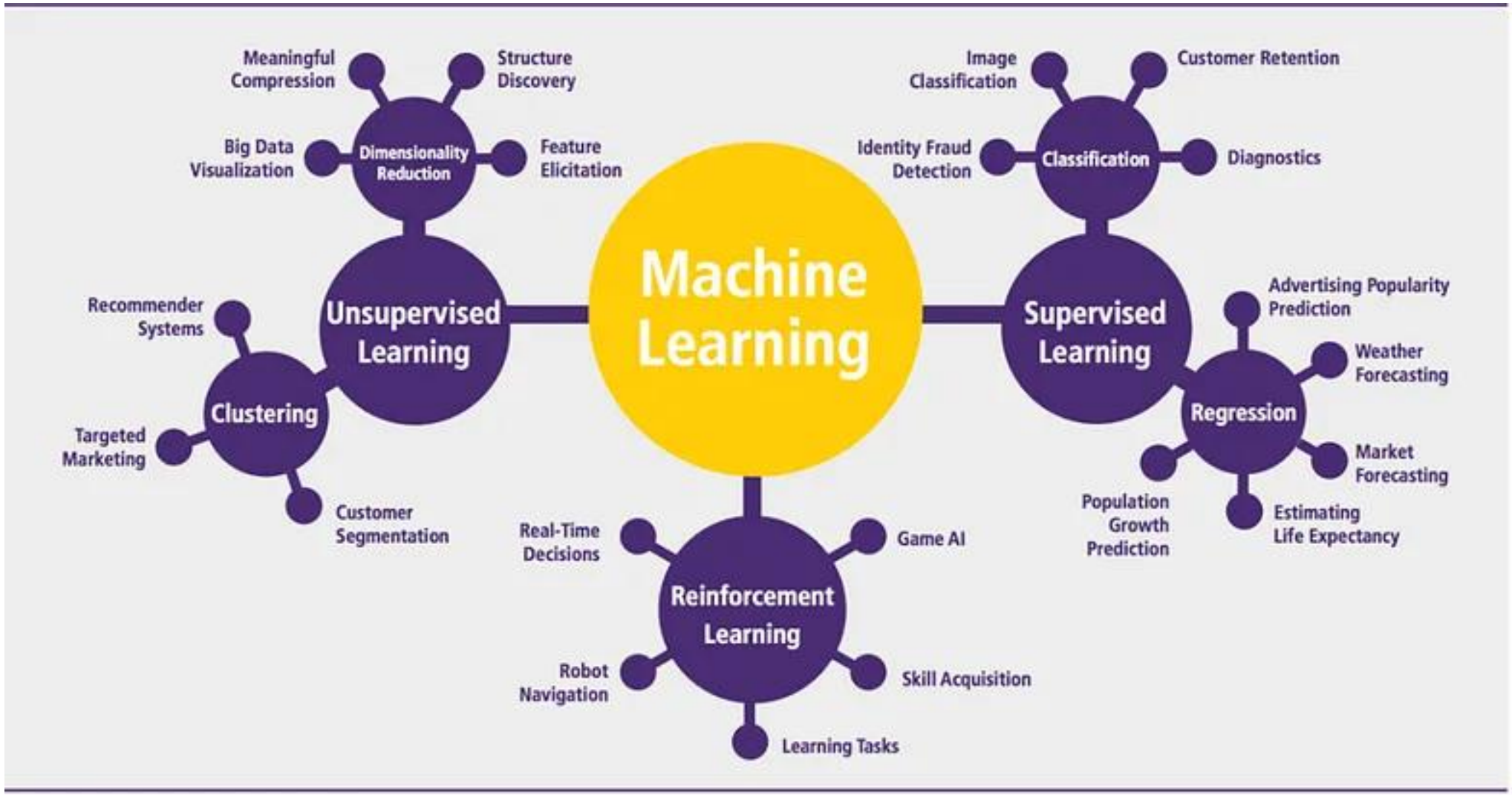
Ce este un LLM?

Elemente AI / Machine Learning:

1. Algoritm / Model inteligent
2. Date
   - Date de intrare
   - Date de ieșire

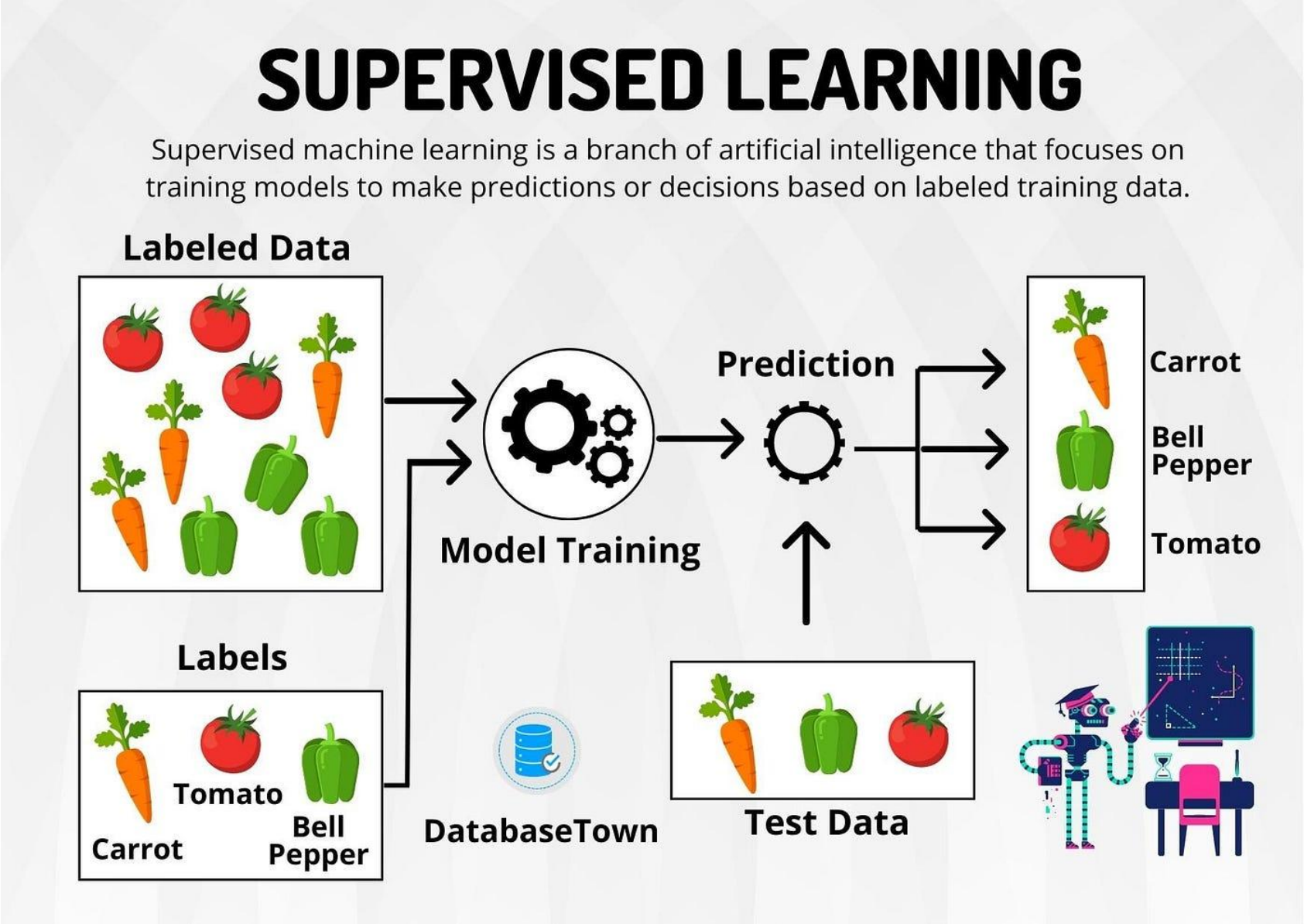


Învățare automată



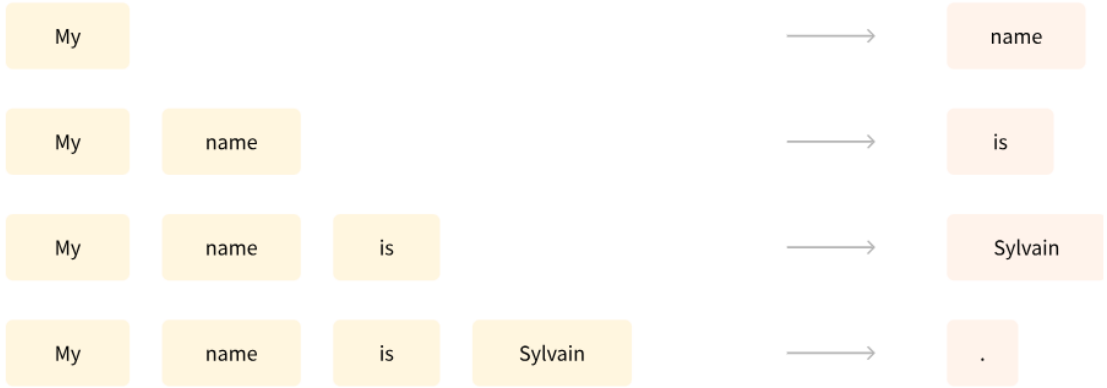
1837 9742

Învățare supervizată

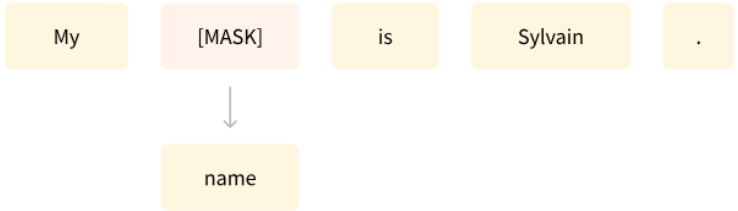


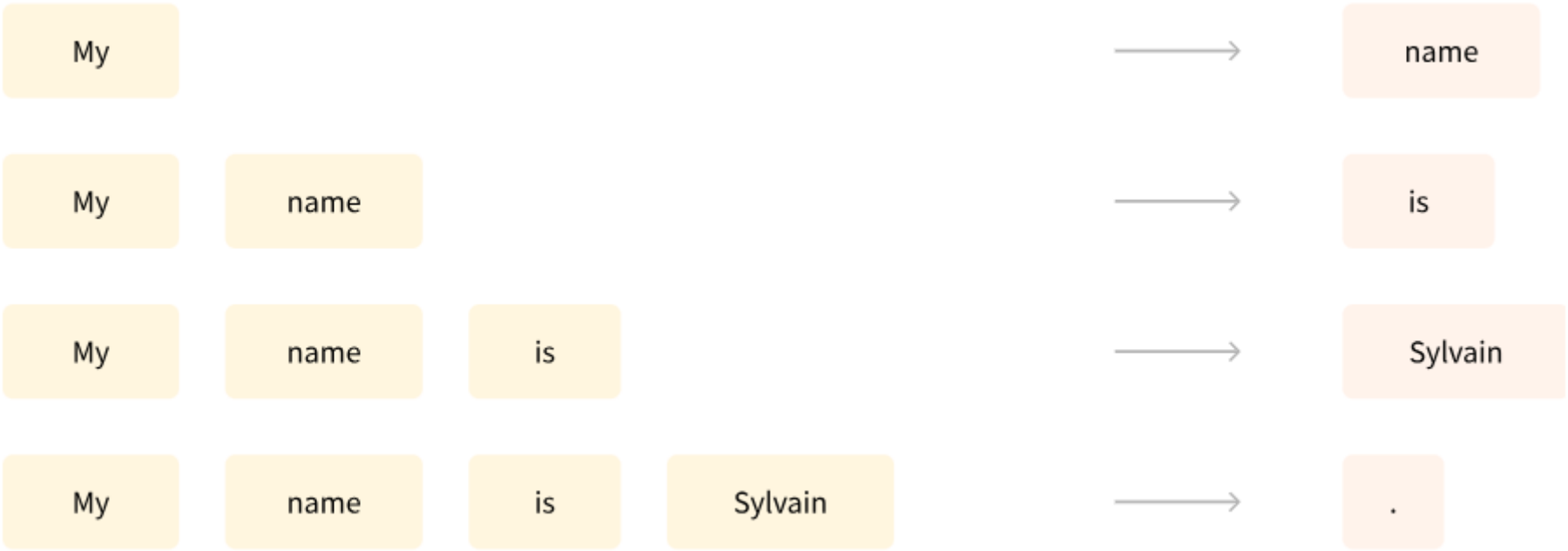
Ce clasifică LLM-urile?

Ce clasifică LLM-urile?

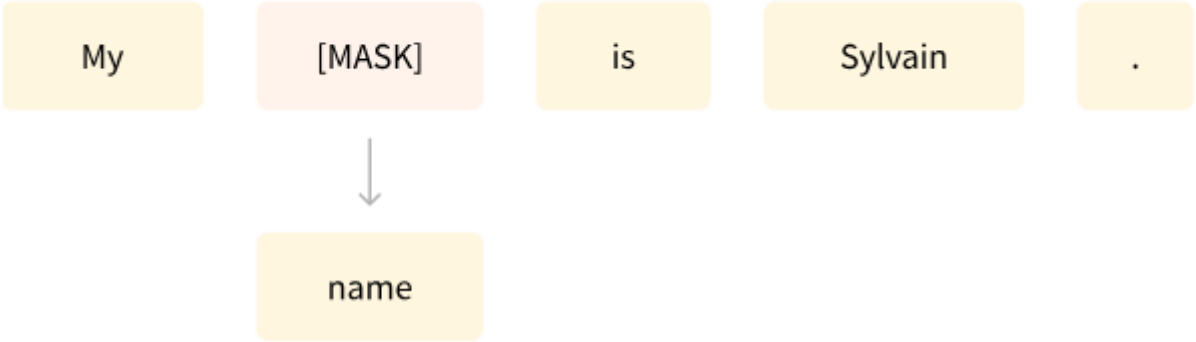


Another example is *masked language modeling*, in which the model predicts a masked word in the sentence.





Another example is *masked language modeling*, in which the model predicts a masked word in the sentence.



Proces de antrenare

Modelele generative sunt

cuv 1 cuv 2 cuv3 cuv4

- Pornind de la un șir de cuvine modelul prezice ce cuvânt ar putea urma

1837 9742

Wild Mentimeter appeared!

Rezultatul predicției

output token probabilities (logits)

model vocabulary size
50,257

0.19850038	aardvark
0.7089803	aarhus
0.46333563	aaron
	...
	...
	...
	...
	...
-0.51006055	zyzzyva

Cuvinte în limba româna?

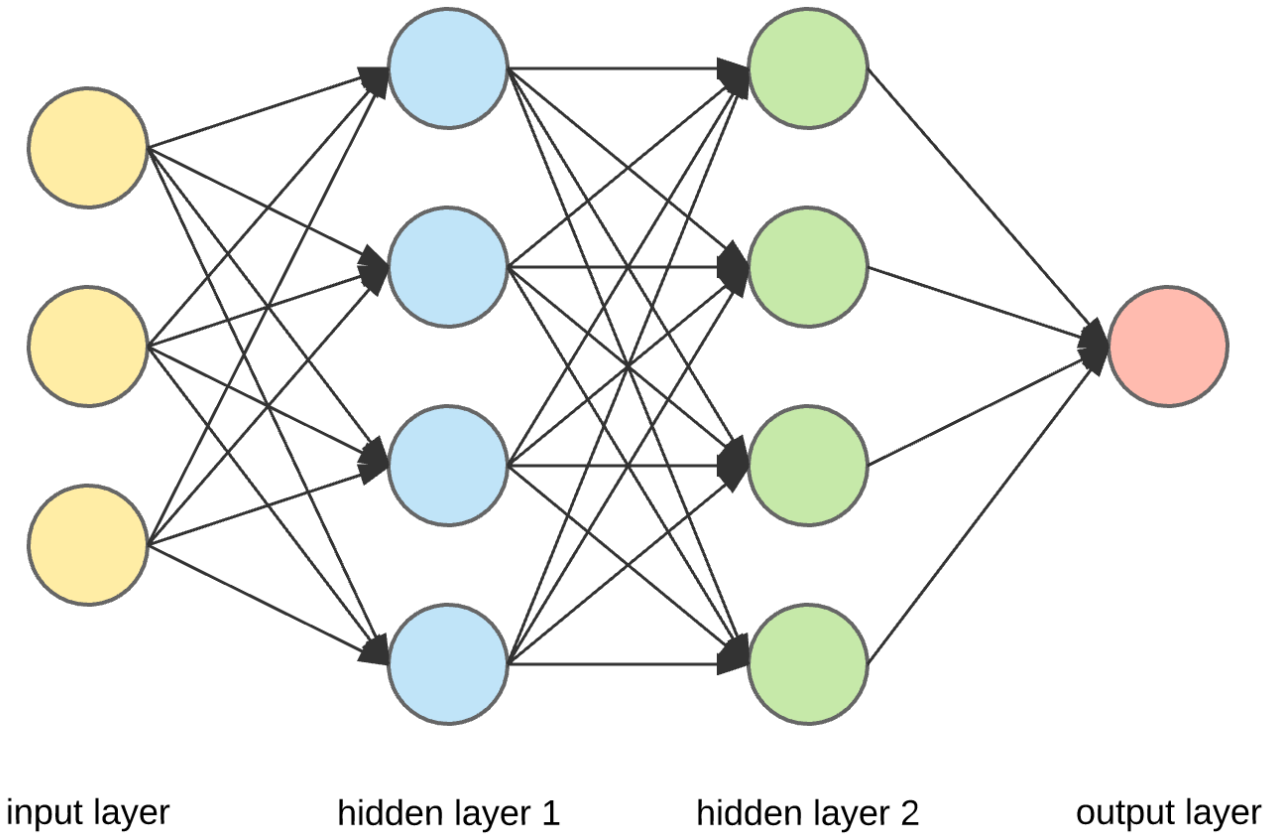
- https://www.menti.com/
- cod: **1837 9742**

Cuvinte în limba româna

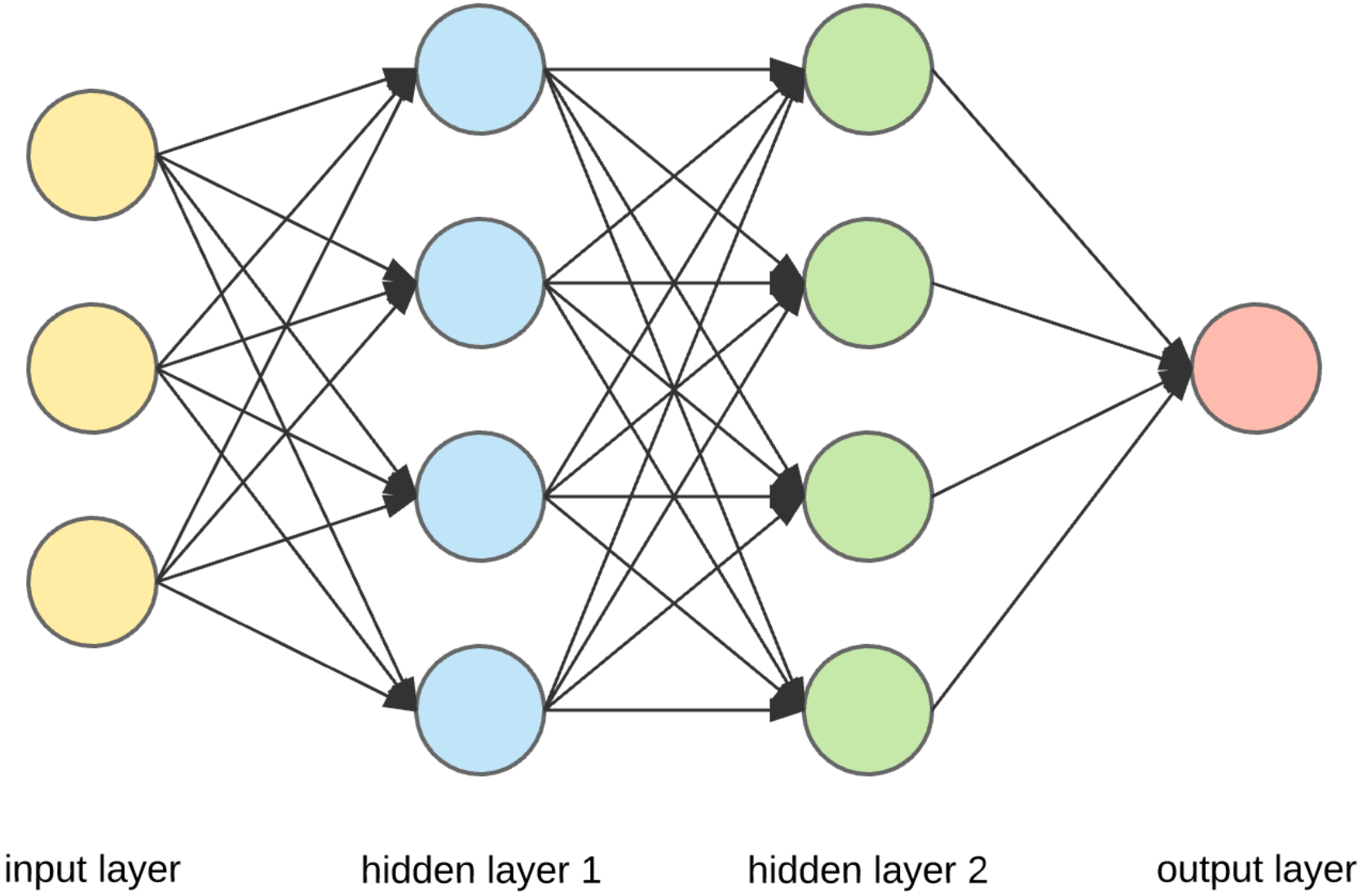
- DOOM – “peste 62.000 cuvinte”
- Dexonline.ro – 75.399 cuvinte
- DLR – 175.000 cuvinte

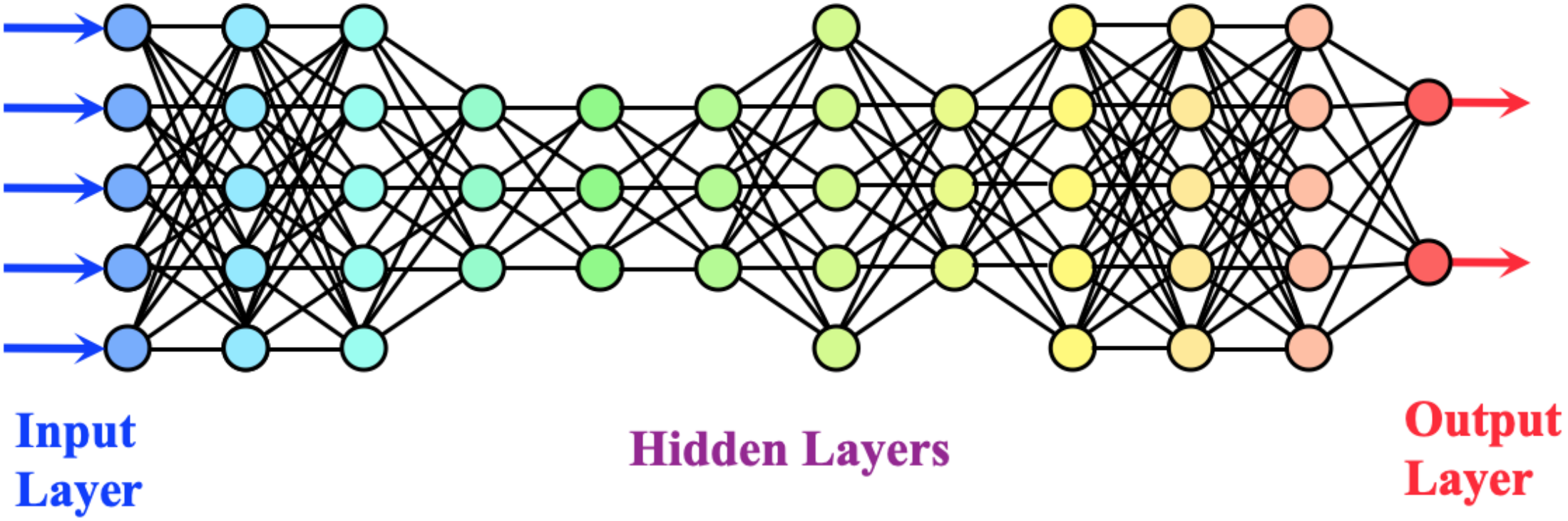
Elemente AI / Machine Learning:

1. Algoritm / Model inteligent
2. Date
   - Date de intrare
   - Date de ieșire

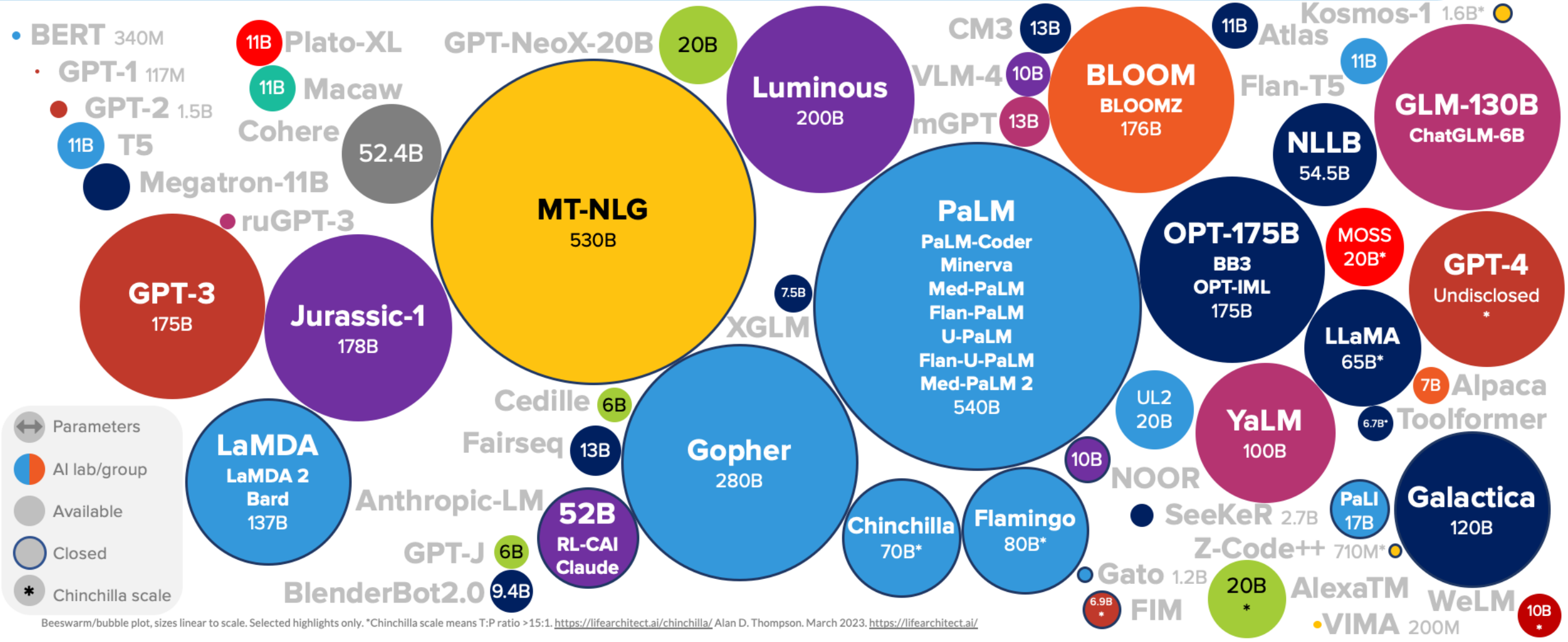


Cum se numește acest algoritm inteligent:
menti.com cod: **5193 0229**




Input
Layer
Hidden Layers
Output
Layer

LANGUAGE MODEL SIZES TO MAR/2023



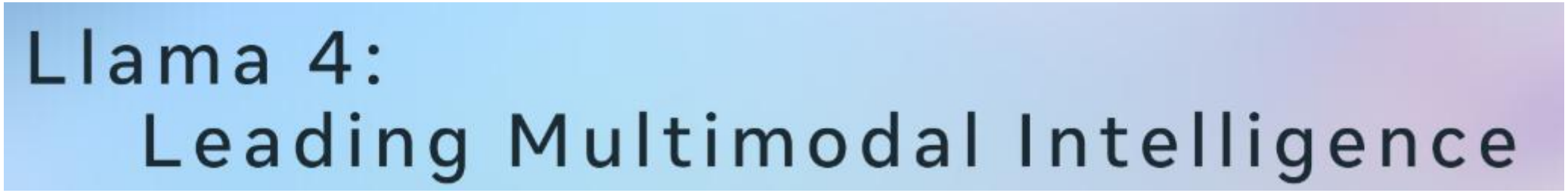
Beeswarm/bubble plot, sizes linear to scale. Selected highlights only. *Chinchilla scale means T:P ratio >15:1. https://lifearchitect.ai/chinchilla/ Alan D. Thompson. March 2023. https://lifearchitect.ai/

Model	Lab	Playground	Parameters (B)	Tokens trained (B)	Ratio Tokens:Params (Chinchilla scaling≥20:1)	ALScore "ALScore" i Sqr Root of	MMLU	MMLU -Pro	GPQA	HLE	Training dataset	Announced ▼	Public?	Paper / Repo	Arch	Tags
AuroraGPT (ScienceGF	Argonne National L	https://lifearchite	*2000*	*30000*	15:1							*TBA*				
DeepSeek-R2	DeepSeek-AI	https://www.reut										*TBA*			*MoE*	Reasoning SOTA
ERNIE 5	Baidu	https://lifearchite										*TBA*				
Grok-4	xAI	https://lifearchite										*TBA*			*MoE*	Reasoning SOTA
GPT-5	OpenAI	https://lifearchite	*5400*	*114000*	22:1							*TBA*			*MoE*	SOTA
GPT-6	OpenAI	https://lifearchite										*TBA*				SOTA
Llama 4 Reasoning	Meta AI	https://ai.meta.cc										*TBA*		https://ai.m	MoE	SOTA Reasoning
~~*MAI-1*~~	~~Microsoft~~	~~https://arstechni~~	~~500~~	~~*10000*~~	~~20:1~~	~~7.5~~						~~*TBA*~~		~~https://www~~	~~*Dense*~~	
o4	OpenAI	https://lifearchite										*TBA*				Reasoning SOTA
Llama 4 Behemoth	Meta AI	https://ai.meta.cc	2000	30000	15:1	25.8		82.2	73.7			Apr/2025		https://ai.m	MoE	SOTA
Llama 4 Maverick	Meta AI	https://ai.meta.cc	400	22000	55:1	9.9		80.5	69.8			Apr/2025		https://ai.m	MoE	SOTA
Qwerky-72B	Featherless AI	https://featherles	72	18000	250:1	3.8	77.46				synthetic, web-scale	Apr/2025		https://featl	Dense	
Agentic-Tx	Google DeepMind	https://github.co	*200*	*20000*	100:1	6.7			62.4	14.5	synthetic, web-scale	Mar/2025		https://stor	*MoE*	
TxGemma	Google DeepMind	https://huggingfa	27	14000	519:1	2.0					synthetic, web-scale	Mar/2025		https://stor	*MoE*	Reasoning
Gemini 2.5 Pro	Google DeepMind	https://aistudio.g	*200*	*20000*	100:1	6.7			84	18.8	synthetic, web-scale	Mar/2025		https://blog	*MoE*	Reasoning SOTA
DeepSeek-V3 0324	DeepSeek-AI	https://chat.deep	685	14800	22:1	10.6		81.2	68.4		synthetic, web-scale	Mar/2025		https://hugg	MoE	SOTA
Llama-3.3-Nemotron-	NVIDIA	https://huggingfa	49	*15040*	307:1	2.9			66.67		web-scale	Mar/2025		https://buil	Dense	Reasoning
EXAONE Deep	LG	https://huggingfa	32	6500	204:1	1.5	83	74	66.1		web-scale	Mar/2025		https://arxi	Dense	Reasoning
Mistral Small 3.1	Mistral	https://huggingfa	24	*8000*	334:1	1.5	81.01	56.03	37.5		web-scale	Mar/2025		https://mist	Dense	
ERNIE 4.5	Baidu	https://yiyan.bai						*79*	*55*		synthetic, web-scale	Mar/2025		https://www	*MoE*	
X1	Baidu	https://yiyan.bai									synthetic, web-scale	Mar/2025		https://www	*MoE*	Reasoning
OLMo 2 32B	Allen AI	https://playgrour	32	6400	200:1	1.5	78				synthetic, web-scale	Mar/2025		https://alle	Dense	
Command A	Cohere	https://dashboar	111	*8000*	73:1	3.1	85				synthetic, web-scale	Mar/2025		https://hugg	Dense	
Gemini Robotics	Google DeepMind		*200*	*20000*	100:1	6.7		*79.1*	*64.7*		synthetic, web-scale	Mar/2025		https://stor	*MoE*	

https://lifearchitect.ai/ - 2025

Exemplu LLM

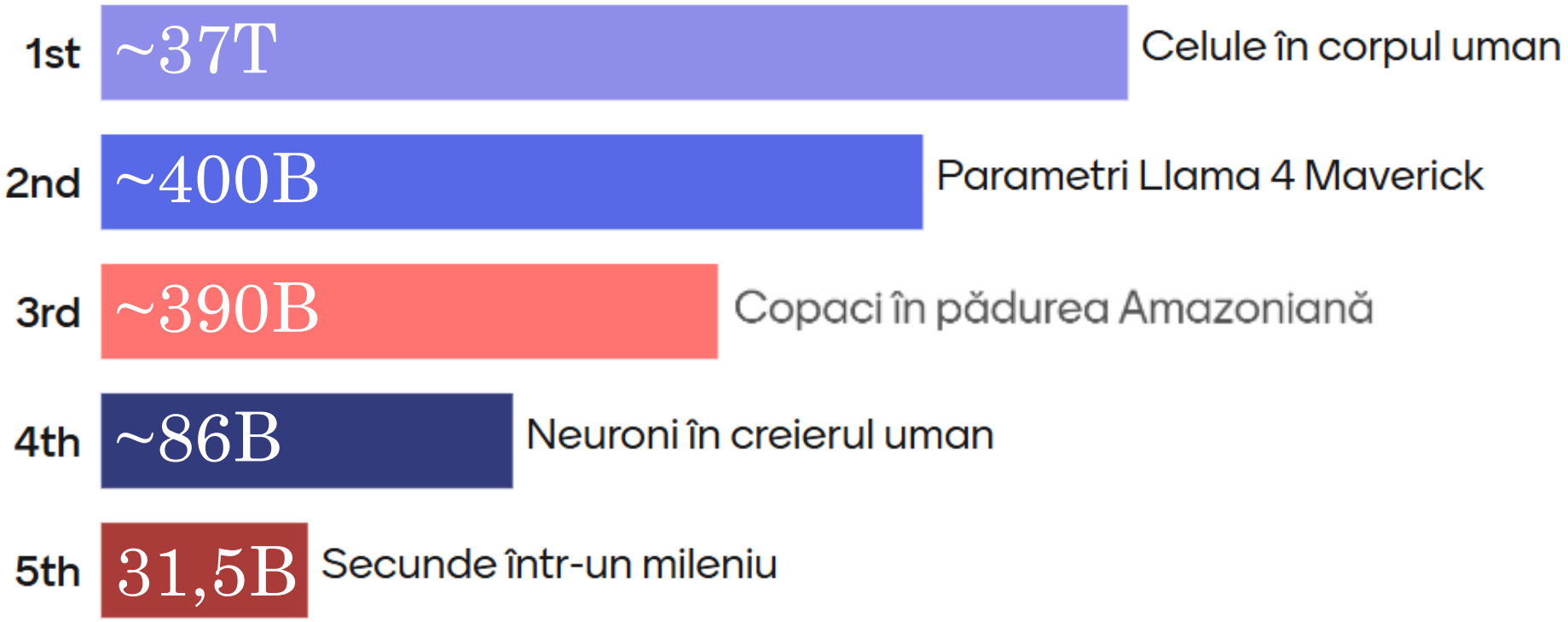
- Llama 4 Maverick
- Dezvoltat de Meta
- 400 **MILIARDE** de parametri



Big numbers

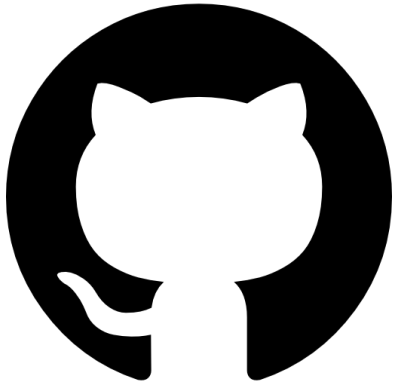
Mentimeter

- https://www.menti.com/
- cod: **5193 0229**


1st
~37T
Celule în corpul uman
2nd
~400B
Parametri Llama 4 Maverick
3rd
~390B
Copaci în pădurea Amazoniană
4th
~86B
Neuroni în creierul uman
5th
31,5B
Secunde într-un mileniu

What is next for Meta?

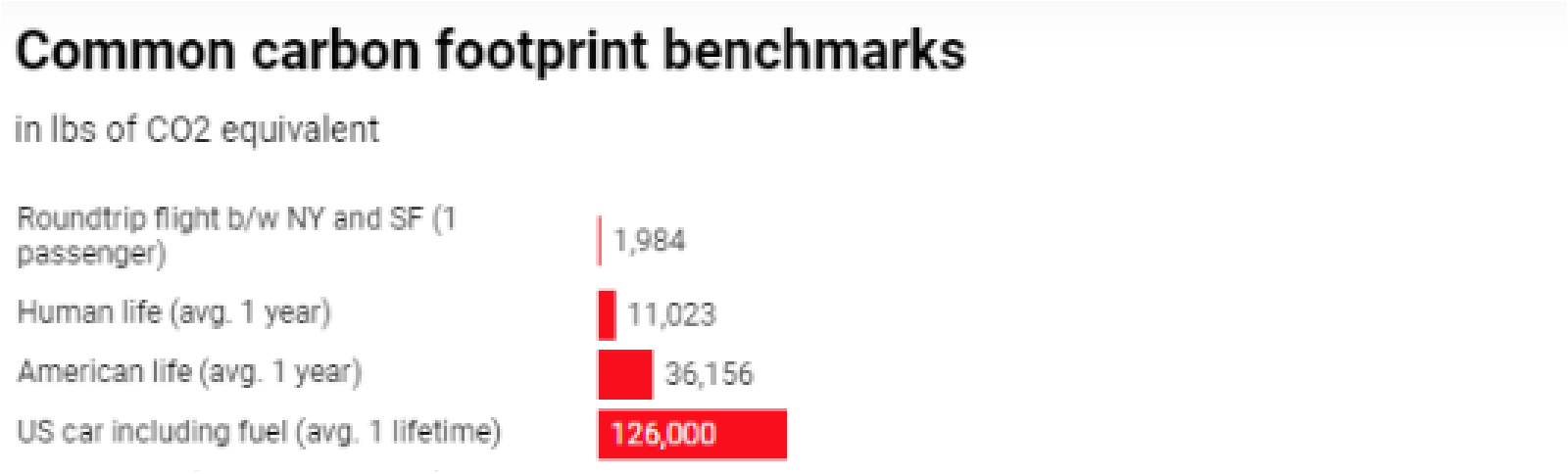
Set de date



Antrenare LLM

- **Pasul 1:** Antrenare inițială (Pre-training)
  - **Scop:** înțelegerea limbajului natural
  - Prezicerea următorului cuvând din text
- **Pasul 2:** Antrenare de bază (Training alignment)
  - **Scop:** dobândirea de competențe specifice
  - Perechi Prompt X + Răspuns așteptat Y
- **Pasul 3:** Reinforcement Learning
  - **Scop:** îmbunătățirea calității umane
  - Ordonarea răspunsurilor + antrenarea unui model capabil să facă singur ordonarea

Probleme Generative AI



Ce urmează în Generative AI? (2025)

- Continuare\Extindere multi-modal LLMs
- Longer memory context
- Autonomous AI Agents
- Reasoning

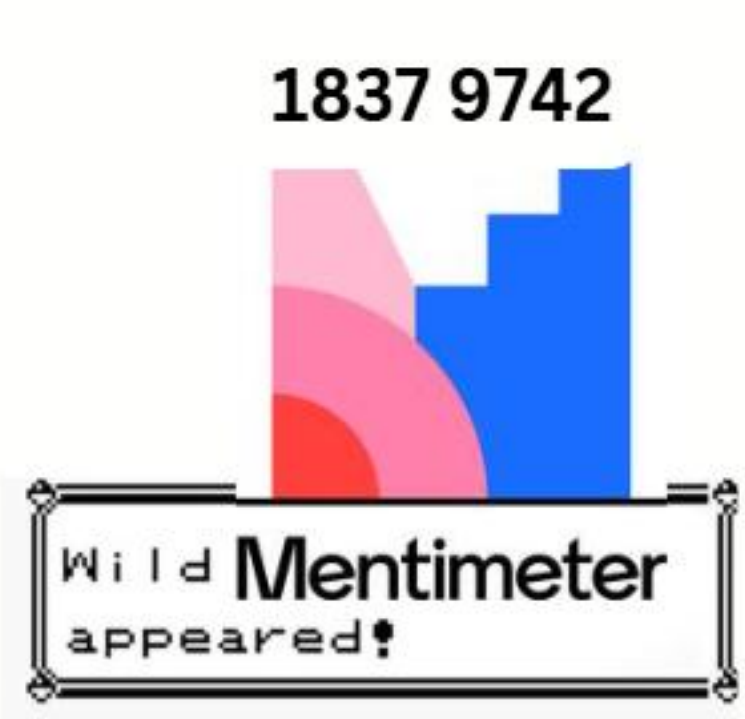
Update 2025

- **Multi-Modal:** real-time voice (audio) + text + image
- **Context:**
  (2025) max ~ 2M tokens
  (2026) max ~ 10M tokens

Update 2025

- **Autonomous AI Agents:**
  - Agent focused development – Antigravity
  - OpenClaw
- **Reasoning**

Ce urmează în Generative AI? (2026)



Resurse utile

- Karpathy Zero to Hero GPT: https://www.youtube.com/watch?v=kCc8FmEb1nY&list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ&index=7
- 3Blue1Brown: https://www.youtube.com/watch?v=wjZofJX0v4M
- Huggin face LLM course: https://huggingface.co/learn/llm-course/chapter1/1
- The annotated Transformer: https://nlp.seas.harvard.edu/annotated-transformer/
- Illustrated GPT2: https://jalammar.github.io/illustrated-gpt2/
- Running a LLM: https://huggingface.co/docs/transformers/llm_tutorial
- Visual Question Answering: https://huggingface.co/docs/transformers/main/en/tasks/visual_question_answering

Sfărșit partea I, II, III

Embeddings

- Embedding projector: https://projector.tensorflow.org/
- Illustrated Word2Vec: https://jalammar.github.io/illustrated-word2vec/
- Word2Vec - StatQuest: https://www.youtube.com/watch?v=viZrOnJclY0
- 3Blue1Brown: https://www.youtube.com/watch?v=eMlx5fFNoYc&list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi&index=8
- Computerphile: https://www.youtube.com/watch?v=gQddtTdmG_8

Mulțumesc!